Three-component decomposition of treatment response in aggregated N-of-1 trials: pharmacological, expectation-driven, and natural-history components

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Abstract

Background. Treatment response in an aggregated N-of-1 trial has three sources: the drug (BR), the participant's belief that treatment is active (PB), and the natural history of the condition (TV). A standard analysis fits one treatment indicator and reports the total. Whether that total misleads depends on the estimand and on whether a candidate biomarker is correlated with the non-pharmacological components. We make that dependence precise.

Methods. We write each component as a modified Gompertz trajectory with participant-specific random effects. We identify which phase contrasts of the hybrid open-label and blinded-discontinuation design separate the components. We derive the omitted-variable identity for the one-component treatment effect and for the one-component biomarker slope. A corollary gives the sign and size of the biomarker-slope displacement from the design's expectancy schedule. Four exploratory Monte Carlo studies, calibrated to the prazosin-PTSD reference parameters, test these predictions. Every reported cell was checked for a positive definite covariance matrix.

Results. With a biomarker correlated with BR alone, the one-component biomarker-treatment estimate is unbiased across a grid of PB and TV magnitudes (Study 1, 1,000 replicates per cell). Its mean absolute bias is 0.016 at $N = 35$ and 0.006 at $N = 300$. A phase-augmented analysis reduces no bias and loses power (0.35 against 0.59 at $N = 35$). When the biomarker is also correlated with PB, the one-component slope is displaced by $-c_{bm,PB}\,\sigma_{PB}\,(\bar{\eta}_{\text{on}} - \bar{\eta}_{\text{off}})$. The sign is set by the design. It is negative on the hybrid design, where belief is higher on drug than off, and positive on a balanced-placebo design, where the reverse holds. The feasible cells of Studies 2 to 4 agree with this prediction in sign and to within about 20 percent in size. Cells of the June 2026 contamination runs at larger biomarker-PB correlations drew from covariance matrices that were not positive definite. Those matrices had been repaired without warning. The apparent bias in those cells is the gap between the nominal and the repaired data-generating process, and it is reported here only as an exhibit of that artifact. The F1 model interacts the biomarker with both drug state and expectancy. On the balanced-placebo design it recovers the pharmacological slope at every feasible contamination level (Study 4, 1,000 replicates per cell).

Conclusions. Decomposition is useful when the estimand is the attribution of response to its sources, or when the biomarker may be correlated with PB or TV. For a biomarker orthogonal to PB and TV, a one-component analysis that uses the design's drug-on/off contrast is sufficient. Recovery of the pharmacological slope under a contaminated biomarker requires a design in which drug exposure and belief vary separately. Both such designs require concealment, and the paper discusses what that entails.

1 Introduction

1.1 A conditional claim about how N-of-1 trials should be analyzed

Treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR, the drug-response component), placebo-belief response (PB, the placebo-response component driven by the patient's belief that they are receiving treatment), and time-variant natural history (TV, the trajectory that would have occurred without intervention). Each is of clinical and regulatory interest, and each is identifiable from a properly designed N-of-1 trial. The standard practice of fitting a single mixed-effects model with one treatment indicator combines the three. We shall argue that whether this combination misleads is conditional. It depends on the estimand and on whether a candi-

date biomarker is correlated with the non-pharmacological components. Where the estimand is the biomarker-treatment interaction and the biomarker is correlated with the drug response alone, the one-component analysis that uses the design's drug-on/off contrast already targets the pharmacological slope. Where the estimand is the attribution of response to its sources, or the biomarker may be correlated with PB or TV, the decomposition is required and the trial design must supply the contrasts that identify it.

We make no claim of novelty for the decomposition itself. The structural decomposition of a longitudinal treatment response into separable disease-progression, placebo, and drug-action components is a well-established tradition in pharmacometrics[1–3]. The separation of drug from placebo response in longitudinal trajectories has been pursued both through growth-mixture models[4] and through experimental decomposition of the placebo response itself[5]. The closest statistical precedent is the linear mixed-model separation of specific from non-specific response in drug-treated subjects by[6], applied by[7] to an open-label fluoxetine phase followed by blinded discontinuation, the design this paper uses. Each ingredient we use, namely mixed-effects models with trial-phase indicators, placebo-arm contrasts, regression-to-mean adjustment, and growth-curve parameterization, has likewise appeared across decades of clinical-trial methodology. Our contribution is therefore one of consolidation and extension rather than invention. We make the three-component structure explicit for the aggregated N-of-1 setting, we add a separately modeled placebo-belief channel identified by design contrasts rather than by functional form alone, and we state the conditions under which the decomposition changes inference. We document why the framing has not been routine in N-of-1 practice, citing parsimony, identifiability concerns, sample-size constraints, and methodological inertia, and we evaluate two linear-mixed-effects approximations to the full decomposition that are tractable at trial-relevant sample sizes.

1.2 Why this matters: three failure modes of the one-component analysis

We begin with the failure modes of the standard approach. A one-component analysis, in which the treatment effect is taken to be what we observe minus baseline, has three identifiable failure modes that arise when the population displays meaningful heterogeneity in any of the unmodeled components. None of these is hypothetical; each has a documented presence in the chronic-condition trial literature. Two points of precision, established formally in section 6.1 and borne out by Study 1 of section 7, should be carried through what follows. The first two failure modes attach to the baseline-referenced one-component effect, and they are removed once the design supplies a drug-on/off contrast, which the one-component analysis of later sections in fact uses. The third failure mode, biomarker-validation failure, is different in kind. It requires not merely that PB or TV be large but that the candidate biomarker correlate with them, and where that correlation is absent the one-component analysis is not biased at all.

The first is the placebo-attribution failure. When the population contains both high and low placebo responders, the one-component treatment effect overestimates the pharmacological component. The bias scales with the magnitude and variance of PB in the study population, and it is invariant under sample-size increases: a larger one-component trial converges

on the same biased estimate. The most consequential example in modern clinical pharmacology is antidepressant efficacy.[8] analyzed FDA-submitted data on four widely-prescribed antidepressants and showed that, except at the most severe baseline-severity stratum, mean drug-placebo differences fell below clinical-significance thresholds, with the apparent 'efficacy' largely explained by placebo response and natural fluctuation.[9] reproduced this pattern in late-life depression and explicitly invoked regression to the mean, and[10] generalized the finding to anxiety. Patient populations differ systematically in placebo responsiveness, and the difference has measurable biological correlates: COMT val158met genotype predicts placebo response in irritable bowel syndrome[11,12], indicating that 'placebo' is not a homogeneous nuisance but a structured biological phenomenon that interacts with the patient's genome and treatment history. A trial whose population happens to be placebo-responsive will report a larger 'drug effect' than one whose population is not, even when the underlying pharmacology is identical.

The second failure mode is regression to the mean. When patients enroll at moments of clinical engagement that coincide with peak symptom severity, natural-history regression toward typical severity contributes several points of apparent improvement in the early weeks of any trial, without any pharmacological contribution at all. The systematic reviews of[13],[14], and[15], the last covering 202 trials across 60 clinical conditions, quantified this directly: in trials with both placebo and no-treatment arms, much of what is conventionally attributed to the placebo response was indistinguishable from regression to the mean and natural-history drift. The one-component analysis absorbs this regression into the treatment-effect estimate, and the analyst has no formal mechanism to recover the true pharmacological effect.

The third failure mode is the biomarker-validation failure. The precision-medicine question that motivates many recent N-of-1 trials is whether a baseline biomarker predicts treatment response, formalized in the PATH statement[16,17]. A biomarker that correlates with placebo responsiveness or with a favorable natural-history trajectory will appear in a one-component analysis as a predictor of the one-component 'treatment effect', indistinguishable from a true pharmacological predictor. The biomarker-stratified prescribing decision that follows from such an analysis is wrong, and the trial's regulatory utility is degraded. The PATH framework explicitly defers the mechanism of treatment-effect heterogeneity to domain knowledge[18,19]; the BR-PB-TV decomposition supplies the mechanistic substrate that PATH leaves blank. We argue that, whenever the candidate biomarker may itself correlate with PB or TV, predictive-biomarker validation cannot be conducted rigorously in a one-component analysis framework regardless of how large the trial is; the bias is fixed by the biomarker's correlation with the non-pharmacological components, not by sample size. Where the biomarker can be assumed orthogonal to PB and TV, this failure mode does not arise, and the one-component analysis recovers the pharmacological slope, as Study 1 of section 7 confirms.

We stress at the outset the precise form this failure mode takes. Study 1 (section 7) varies the magnitude of PB and TV with the biomarker correlated with BR alone. The section 6.1 identity predicts no bias there, and none appears. Studies 2 to 4 correlate the biomarker with PB as well. The identity then predicts a displacement of the one-component slope equal to $-c_{bm,PB}\,\sigma_{PB}\,(\bar{\eta}_{\text{on}} - \bar{\eta}_{\text{off}})$, where the bars are the design's mean expectancy over on-drug and

off-drug visits. The operative quantity is the biomarker's correlation with the individual-level PB amplitude ($c_{bm,PB}\,\sigma_{PB}$), not the population-mean placebo response m_{PB}, which does not enter the slope. A trial population can have a large average placebo response and yet pose no validation hazard, while one with zero average but a biomarker-correlated spread of placebo responsiveness is at risk. The sign of the displacement is a property of the design. The feasible simulated cells agree with the prediction. Study 4 shows that the F1 model, which interacts the biomarker with both drug state and expectancy, recovers the pharmacological slope on a design that varies drug exposure and belief separately.

1.3 The current state of the N-of-1 literature

The N-of-1 trial in its modern randomized form originates with[20], who defined the within-patient multiple-crossover RCT, demonstrated it on a single airflow-limitation case, and established a clinical service to deliver it at scale. The design inherits its identifying logic from two older traditions: the crossover trial of classical biostatistics, in which within-subject period contrasts estimate treatment effects under explicit carryover assumptions, and the single-case experimental designs of behavioral science, in which phase reversals (withdrawal and reinstatement of treatment) identify effects within one participant. The blinded-discontinuation contrast central to this paper is a direct descendant of both. The methodological step from the individual-patient trial to population-level inference was taken by[21] and[22], who showed that a series of single-patient trials can be combined through hierarchical meta-analysis to estimate both population treatment effects and individual responses, the statistical foundation on which the aggregated design rests. Over the last fifteen years the design has accordingly been reframed from an individual-patient decision tool into an aggregated, meta-analytic framework supporting population-level inference: the programmatic statement of[23], the methodological reviews of[24], the historical-practice audit of[25], and the more recent syntheses of[26],[27],[28], and[29] trace that trajectory. Reporting standards are codified in the CONSORT extension[30] (the headline statement) and[31] (the explanation-and-elaboration companion); the reporting-quality audits of[32] and its successors document slow but progressive improvement in analytic transparency. Three observations about this literature motivate the present paper.

The first observation is that the dominant analytic models are simple. The reporting-quality audit of[32] documents that the published N-of-1 trials between 1985 and 2013 relied predominantly on paired t-tests, simple linear mixed-effects models with a binary treatment indicator, or hierarchical Bayesian regressions. On our own reading of that corpus and the methodological literature that followed it, no published N-of-1 trial has employed a structural decomposition of treatment response into pharmacological, expectancy, and natural-history components. The most recent methodological advances retain this character:[33] develops Bayesian distributed-lag models with autocorrelated errors as a current carryover-aware analysis, but the model still treats the response as a single quantity to be estimated.[34] and[35] compare aggregated N-of-1 trials with parallel and crossover RCT alternatives by simulation and document carryover-induced Type I error inflation as a key failure mode, but again at the level of the one-component treatment effect. None of these papers decompose the response.

The second observation is that the placebo-component separation problem is widely known

but largely unaddressed in N-of-1 specifically. Outside the N-of-1 setting,[6] and[7] separate specific from non-specific response in drug-treated subjects with linear mixed models and prototype outcome profiles, using an open-label phase followed by blinded discontinuation. The blinded discontinuation phase used in the Hendrickson hybrid design[36] is one of the few protocol-level mechanisms in routine N-of-1 use that supplies identifiability of PB versus BR. Other relevant approaches, including placebo run-in designs, open-label-placebo trials, and mediator-analysis applications, have been developed in parallel but rarely imported into N-of-1 methodology.

The third observation is that the closest precedent in the literature is Kaptchuk's three-component decomposition of the placebo response itself. In a randomized trial of acupuncture-style placebo for irritable bowel syndrome,[5] decomposed the placebo response into observation, ritual, and patient-practitioner relationship components, demonstrating empirically that the placebo response is itself a sum of identifiable parts rather than a single 'expectancy' lump. The follow-up open-label placebo trials of[37],[38], and[39] in adolescent functional pain establish that even disclosed-placebo intervention can produce clinically meaningful PB-only response. The framework we propose is structurally analogous to Kaptchuk's decomposition but applied at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone. To the best of our knowledge, no published paper proposes the corresponding decomposition for the aggregated N-of-1 setting, even though the methodological ingredients, including mixed-effects modeling, blinded-phase contrasts, and trajectory parameterization, are all in routine use.

The case we develop in the rest of this paper is therefore that the gap is real, that the methodological tools to fill it are mature, and that the conditions under which the one-component analysis fails can be stated exactly and checked by design.

1.4 The neurobiological and epidemiological reality of the components

The decomposition is not a statistical convenience. Each component reflects a real causal mechanism documented in independent literatures, and we shall briefly recall each in turn before turning to the formal model.

The biological response component represents the pharmacological action of the active compound: the chemical binding, downstream signaling, and physiological response that a regulatory authority cares about when it grants approval. The placebo-belief response reflects neurobiologically real mechanisms documented across pain, depression, Parkinson's, and PTSD research[40–43], mediated by belief-driven release of endogenous opioids and dopamine, anticipatory regulation of the hypothalamic-pituitary-adrenal axis, and top-down attentional shifts in symptom perception. Direct neuroimaging evidence is instructive here:[44] developed an fMRI-based 'neurologic pain signature' (NPS) that tracks nociceptive input and is reduced by the opioid agonist remifentanil, and the individual-participant meta-analysis of[45] found that placebo treatments produce moderate reductions in reported pain but only negligible reductions in the NPS, providing a neural dissociation between the pharmacological and expectancy contributions to a clinical pain response. The placebo response is heterogeneous across pa-

tients with biological correlates[11,12] and is modulated by trial phase: when belief varies, as it does between open-label and blinded-discontinuation conditions, the placebo response varies, and the modulation is exactly the design contrast that makes the decomposition identifiable.

The time-variant natural-history component reflects whatever trajectory the participant would have followed without treatment. For acute conditions this trends positive, that is, recovery on an intrinsic timescale; for chronic stable conditions it is near zero on the population mean but with substantial participant-level variance; for chronic progressive conditions it trends negative; and for selection-driven enrollment processes it is positive in the early weeks of the trial regardless of underlying disease class, because of regression to the mean[13,15]. The PATH statement[16,17] highlights that this heterogeneity is itself the central object of precision-medicine inference: HTE must be modeled, not absorbed into noise, if the precision-medicine question is to be answered.

1.5 What this paper contributes

We make four contributions, each addressed in a section below. We begin in section 2 with the formal model and notation. Section 3 develops each component in turn, with biological motivation, parametric form, and identifiability conditions. Section 4 maps the design-contrast structure of the standard hybrid open-label-plus-blinded-discontinuation design onto the identifiability requirements of the decomposition. Section 5 develops the analysis-side methodology, including the linear-mixed-effects approximations and the phase-by-treatment indicator extension that retains BR-PB sensitivity at low parametric cost. Section 6 presents the worked examples, and section 7 presents the simulation study. Section 8 discusses applications to predictive-biomarker validation. The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` provides an extended worked-example treatment for readers approaching the material for the first time. A glossary of project terms precedes the references.

2 Notation and the formal model

In this section we shall fix notation and state the formal model. The three components are defined precisely enough to make the identifiability and analysis-side arguments in later sections rigorous, but we have endeavored to keep the presentation compact. A fuller treatment, with worked numerical examples at each step, is available in the companion pedagogical document at `docs/24-component-decomposition-pedagogy.md`.

For participant i at trial timepoint t, we may write:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[\, BR_{it} + PB_{it} + TV_{it} \,\right] \;+\; \varepsilon_{it}.$$

Each component is a participant-specific function of its own time-on-state variable. BR_{it} depends on time-on-drug, the cumulative exposure to active medication since drug initiation.

Table 1: Principal terms used in the three-component decomposition. The distinction between D_{it} and $D_{bc,it}$ follows the project notation.

Term	Meaning
BR	Drug-response component: pharmacological reduction in symptom score attributable to active drug
PB	Placebo-response component: reduction attributable to the patient's belief that they are receiving t
TV	Natural-history component: reduction (or increase) attributable to the trajectory of the underlying
η_t	Expectancy at visit t, the multiplier on the PB mean and standard deviation: 1 when the participa
BL_i	Participant i's baseline (pre-treatment) symptom score.
D_{it}	Binary drug state: 1 on drug, 0 off drug.
$D_{bc,it}$	Continuous exposure-decayed drug indicator on the analysis side: 1 on drug, $\exp(-\lambda t_{sd,it})$ off drug
$t_{sd,it}$	Time since drug discontinuation (off-drug only).
λ	Carryover decay rate, with half-life $t_{1/2} = \ln 2/\lambda$; 1 week in every simulation here.
c_{bm}	Correlation between the biomarker and the drug-response component on drug (0.45 in every alterna
$c_{bm,PB}$	Correlation between the biomarker and the placebo-response component at every visit (the contami
σ_{PB}	Standard deviation of the participant-level PB amplitude at $\eta = 1$ (2 points here).

PB_{it} depends on time-on-belief, modulated by a phase-specific expectancy factor η(phase) that captures the patient's confidence that they are receiving active treatment. TV_{it} depends on time-since-trial-entry, calendar time elapsed regardless of treatment status.

Each component follows a modified Gompertz curve. In symbols:

$$C(t) \;=\; m_C \exp(-d_C \exp(-r_C t)) \;-\; m_C \exp(-d_C),$$

where m_C is the scale (the asymptote up to the factor $1 - e^{-d_C}$), d_C the displacement that sets the initial lag, and r_C the onset rate, each discussed for BR in section 3.1. ε_{it} is a within-participant residual with first-order autoregressive correlation in calendar time, $\mathrm{Cov}(\varepsilon_{is}, \varepsilon_{it}) = \sigma_\varepsilon^2 \rho^{|w_t - w_s|}$ with w_t the week of visit t. The three components are jointly correlated: c_1 is the between-component correlation at the same visit and $c_\times$ the between-component correlation across visits, so that participant-level heterogeneity in one component can co-vary with heterogeneity in the others. The simulations fix $\rho = 0.7$, $c_1 = 0.1$ and $c_\times = 0.05$.

The principal psychometric and pharmacological terms used below are summarized in Table 1.

3 The three components

In this section we shall present the three components in turn, each with its biological motivation, the modified Gompertz parameterization, and the design contrasts that identify it. The treatment is organized so that the reader can see, by the end of this section, that the formal model is a sum of three causally interpretable trajectories rather than a mathematical convenience.

3.1 Biological response (BR)

BR is the physiological reduction in symptom score attributable to the active medication's chemical action on the body. It is independent of belief and natural history: a participant who somehow received active drug without knowing it would still develop a measurable BR, and a participant who took an inert sugar pill while believing it was active drug would have $BR = 0$ no matter how strong their expectation. Methodologically, BR is the quantity that drug regulators and prescribers want: approval decisions, dose selection, and biomarker-guided patient stratification are all framed around BR.

The qualitative shape of the BR trajectory follows from pharmacokinetic and pharmacodynamic considerations. Drug concentration in plasma reaches steady state within a small multiple of the elimination half-life, and brain concentration follows plasma with a short delay. The downstream pharmacodynamic response, that is, the reduction in symptom-relevant neural activity in the case of prazosin's effect on noradrenergic tone during REM sleep, typically requires several days to weeks of sustained exposure to integrate fully[46]. The result is a saturating curve: slow rise in the first week, steepest growth in the second to third week, plateau as receptor adaptation completes, and a decay back toward zero on discontinuation governed by the carryover half-life $t_{1/2} = \ln 2/\lambda$ rather than an instantaneous drop. The rate of this off-drug decay is the object of the carryover machinery used throughout this project; only in the limit of a short half-life relative to the measurement spacing does it approximate the sharp step assumed in the worked example of section 6.

The modified Gompertz form $BR(t) = m_{BR} \exp(-d_{BR} \exp(-r_{BR} t)) - m_{BR} \exp(-d_{BR})$ captures this profile compactly. The asymptote $m_{BR}(1 - e^{-d_{BR}})$ is the biological maximum, the largest reduction the drug can produce at infinite exposure. The scale parameter m_{BR} equals that maximum up to the factor $(1 - e^{-d_{BR}})$, which for the calibrated $d_{BR} \approx 5$ exceeds 0.99. The onset rate r_{BR} controls how rapidly the participant approaches the maximum: high r_{BR} means a fast onset (effect plateaus in one to two weeks), low r_{BR} a slow climb (six to eight weeks). The displacement d_{BR} sets the length of the initial lag: high d_{BR} gives a sigmoid curve with a slow start and an inflection at $\ln d_{BR}/r_{BR}$, later for larger d_{BR}, and the maximum slope decreases as d_{BR} grows; low d_{BR} ($d_{BR} \leq 1$) gives a curve with no lag that rises fastest at the start (`analysis/report/compendium-mathematics.tex`, Section "Shipped parameters, landmarks, the displacement, and numerical conditioning' '). For PTSD prazosin, calibrated values from the[36] reference dataset are $m_{BR} \approx 11$ (eleven points of nightmare-score reduction at saturation), $r_{BR} \approx 0.42$ per week (half-maximum at roughly four to five weeks of sustained exposure), and $d_{BR} \approx 5$ (a moderate initial lag, with the inflection at about 3.8 weeks).

Two alternative families that do not capture the asymmetric saturation profile are worth noting for contrast. A linear function $BR = \beta t$ climbs without bound and is biologically implausible past a few weeks. An exponential decay $BR = m_{BR}(1 - \exp(-rt))$ is acceptable for fast-onset drugs, but it lacks the slow-start phase that characterizes most chronic-condition responses, where receptor adaptation, feedback regulation, and gene-expression changes produce a measurable lag. The Gompertz interpolates between exponential decay (high d limit) and linear-in-t growth (low d limit), and its three-parameter form is flexible enough to fit a

wide range of pharmacodynamic profiles without overfitting.

The Gompertz is a deliberate choice over the two parametric forms most familiar from pharmacokinetic-pharmacodynamic modeling. The sigmoid $E_{\max}$ (Hill) model[47] describes the dependence of effect on concentration, that is, a dose-response curve, whereas $BR(t)$ here is a time-course of accumulating effect at fixed dosing; and the indirect-response (turnover) models that do describe effect time-courses are specified through differential equations on a hidden response variable rather than in closed form. The modified Gompertz retains the slow-start, saturating shape those models produce while remaining a closed-form, three-parameter mean function that is tractable inside a mixed-effects likelihood. Doc 25 of the project documentation provides the full historical and biological context for the family choice and quantifies the sensitivity of pmsimstats inference to alternative families; that work is reported separately as `07-gompertz-evaluation`. The headline result of that evaluation is reassuring for the present framework: in a 1{,}000-replicate-per-cell simulation that generated data under the logistic, hyperbolic-tangent, and piecewise-linear breakpoint families and analyzed them under the standard linear-mixed model, the trajectory family was essentially irrelevant to the power of the biomarker-treatment interaction test (the family-to-family spread fell within two MCSEs of sampling noise) and to Type I error (0.025 to 0.036 across null cells, with no family-specific inflation). The conclusions drawn here are therefore not artifacts of the Gompertz form, though the regimes the family cannot represent, namely cyclical or biphasic trajectories, remain genuine exclusions.

The BR component is identified by within-participant on-drug versus off-drug contrasts, formalized in section 4 below. The most direct contrast is the blinded-discontinuation phase: when a participant is silently switched to placebo while still believing they are on active drug, BR decays toward zero as the drug clears while PB remains roughly intact, and the difference between on-drug and off-drug response in that window is a relatively clean estimate of BR that does not require the analyst to model PB or TV explicitly[7]. The decay is not instantaneous: with the carryover half-life $t_{1/2} = \ln 2/\lambda$ of the analysis-side decay term, residual BR persists into the early off-drug observations, so the contrast is clean only once the discontinuation window is long relative to $t_{1/2}$, and the drug state is best carried by the continuous-decay indicator $D_{bc,it}$ of Table 1 rather than the binary state D_{it}.

3.2 Placebo-belief response (PB)

PB is the reduction in symptom score that arises from the patient's belief that they are receiving active treatment, independent of what they are actually receiving. That is, it is the placebo response in its strict sense: the part of the improvement that would persist if the active drug were silently replaced with an inert pill, provided the patient continued to believe they were on the active drug.

It is worth being precise about what PB is and is not. PB is not 'fake' improvement; it is a phenomenon clinicians knowingly exploit, and a national survey of US internists and rheumatologists found roughly half report regular use of placebo treatments[48]. As the introduction sets out, the placebo response is mediated by well-characterized neurobiologi-

cal pathways[40–43,49]: belief-driven release of endogenous opioids and dopamine, anticipatory hypothalamic-pituitary-adrenal regulation, and top-down attentional shifts, dissociated from pharmacology by the[44] neurologic pain signature[45] and shown to have genetic correlates[11,12]. What makes these mechanisms 'placebo' rather than 'drug' is that they are triggered by belief rather than by pharmacology.

For trial design, the relevant point is that PB is separately identifiable from BR if and only if the trial includes a phase in which the patient's belief about treatment differs from the actual treatment they are receiving. The blinded-discontinuation phase is the canonical such phase: the patient is told that, at some unannounced point during the window, they may be silently switched to placebo, and they cannot reliably tell whether they are still receiving active drug. Belief in treatment correspondingly drops to a partial level (multiplier roughly 0.5) because the patient's expectation is hedged.

PB also follows a Gompertz curve, scaled by the expectancy factor. In symbols:

$$PB(t \mid \text{phase}) \;=\; \eta(\text{phase}) \cdot m_{PB} \exp(-d_{PB} \exp(-r_{PB} t)) \;+\; (\text{offset to start at zero}).$$

The expectancy values used in this paper are those of Table 1: $\eta = 1$ when the participant knows treatment is active, $\eta = 0.5$ in blinded discontinuation, $\eta = 0$ when active drug is given covertly, and $\eta = 1$ when placebo is substituted while the participant is told treatment continues. The last two phases belong to the balanced-placebo design of Study 4 and both require concealment, discussed in section 7. A disclosed open-label placebo, in which the patient is told the pill is inert, is a different intervention with an expectancy between 0 and 0.5; it is not simulated here. The quantity η operationalizes response expectancy in the sense of[50], the patient's anticipation of benefit that the expectancy literature treats as a determinant of experienced symptom change. The choice of $\eta = 0.5$ for the blinded phase is a modeling assumption rather than a calibrated quantity: to our knowledge the residual expectancy a patient carries when uncertain about allocation has not been measured directly as a fraction of the open-label level, and we adopt 0.5 as a deliberately central placeholder. The available indirect evidence is nonetheless that the residual response is substantial. Open-label placebo trials[37–39] demonstrate that a clinically meaningful response persists even when patients are explicitly told they are receiving an inert intervention, which bounds the residual belief-driven response away from zero. Because the value of η is not empirically pinned, a sensitivity analysis varying it across its plausible range is needed before any conclusion is allowed to depend on this single number. That analysis was not run for this paper (section 7.8).

The decomposition framework as proposed here has a structural analogue in the placebo-research literature.[5] decomposed the placebo response itself into observation, ritual, and patient-practitioner relationship components in a randomized trial of acupuncture-style placebo for irritable bowel syndrome, demonstrating empirically that the placebo response is not a single 'expectancy' lump but a sum of identifiable parts. The present BR-PB-TV framework is analogous in spirit but operates at the level of the full treatment response (drug, placebo, natural history) rather than within the placebo response alone.

A subtle property of PB is that its variance also scales with η, not just its mean. When η is high, there is substantial variability across patients in how strongly the placebo response is expressed; high-expectancy patients show large PB, and low-expectancy patients show essentially none. When η is low (blinded), the placebo response is uniformly attenuated and the between-participant variance shrinks accordingly. The model therefore parameterizes $\mathrm{SD}(PB \mid \text{phase}) = \eta(\text{phase}) \cdot \sigma_{PB}^{\text{baseline}}$. Ignoring this heteroscedastic structure deflates standard errors in the open-label phase and inflates them in the blinded phase, with downstream consequences for confidence-interval coverage and for the calibration of any biomarker test built on top of the analysis.

3.2.1 Additivity, subadditivity, and the BR-PB interaction

In this section we shall broach a question that the formal model leaves open: whether the three components contribute additively to symptom reduction, or whether there is a meaningful interaction between BR and PB that the additive specification misses.

The decomposition as written in section 2 treats the three components as additive contributors to symptom reduction. In symbols:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; (BR_{it} + PB_{it} + TV_{it}) \;+\; \varepsilon_{it}.$$

The additivity is a modeling assumption, not a fact about biology, and it is worth stating that the assumption may be wrong in directions the framework can absorb and in directions it cannot. Additivity is also scale-dependent in the sense emphasized by[51]: components that combine non-additively on the raw symptom scale may combine additively on a transformed (for example logarithmic) scale, so an apparent $BR \times PB$ interaction can be partly an artifact of the chosen response metric rather than a biological fact. That belief and pharmacology interact rather than act in isolation is demonstrated directly by[52], who showed with neuroimaging that positive treatment expectation roughly doubled the analgesic benefit of the opioid remifentanil while negative expectation abolished it, so the two channels are not in general separable by simple addition. The same study is the direct evidence that the combined effect of receiving an active drug and believing one is receiving it can differ from the sum of the two measured separately;[40] reviews the neurobiological basis. When the combined effect falls below the sum the phenomenon is termed subadditivity: $\partial Y / \partial (BR \cdot PB) > 0$ when both components are present and their interaction term is included in a saturated mean-structure model.

Three mechanisms have been proposed in the placebo-research literature to explain subadditivity. First, a shared neural substrate: if the descending pain-modulation circuits that mediate placebo analgesia overlap with the circuits modulated by the active drug, simultaneous activation by both mechanisms produces less marginal benefit than activation by either alone, in a saturating-occupancy sense. Second, the bound of the symptom scale: a clinical scale that caps at zero (no symptoms) cannot register additional reduction once a participant has reached that cap, so any participant whose $BR + PB$ already saturates the

scale will register no further reduction even if BR or PB would, in isolation, predict more. Third, belief-dynamics adaptation: the patient's expectation may adapt downward when the active drug produces a strong response that exceeds the expected effect, reducing the ongoing PB contribution; equivalently, unexpectedly large BR may attenuate PB via expectation recalibration.

Whichever mechanism dominates in a given clinical context, the subadditivity question is empirical and design-dependent. Within the BR-PB-TV framework, subadditivity manifests as a non-zero coefficient on a $BR \times PB$ interaction term in the conditional-mean specification. The additive form treats this interaction as zero by construction; a generalized form, in symbols:

$$Y_{it} \;=\; \mathrm{BL}_i \;-\; \left[BR_{it} + PB_{it} + TV_{it} + \gamma \cdot BR_{it} \cdot PB_{it}\right] \;+\; \varepsilon_{it},$$

For trial design, two consequences follow. First, a biomarker that predicts BR in an additive-decomposition analysis may overstate or understate predictive accuracy if the true mechanism is subadditive: under subadditivity, the biomarker-driven increase in BR is partly cancelled by the $\gamma \cdot BR \cdot PB$ interaction, and the marginal biomarker-treatment association is smaller than the biomarker-by- BR association. Second, trial designs that attenuate PB in some phases (blinded discontinuation, crossover) implicitly down-weight the subadditivity contribution to the response in those phases, which means that the effective biomarker-treatment association estimated from those phases is closer to the biomarker-by-BR association than is the open-label estimate. The phase-augmented analysis recommended in section 5 below therefore has the side benefit of partly absorbing subadditivity, even when γ is not estimated explicitly.

A simulation quantifying the bias in the biomarker-treatment interaction estimate when the additive analysis is fitted to data generated under non-zero γ was specified in the working protocol as Study C but was not run for this paper (section 7.8). The working hypothesis is that bias scales linearly with γ at small $|\gamma|$, with the sign opposite to the true biomarker-by-BR effect under subadditivity and aligned with it under superadditivity.

A subtle but inferentially important point merits attention here: a non-zero $\hat{\gamma}$ recovered from an additive-extension analysis does not, on its own, establish that the underlying mechanism is subadditive interaction in the strict sense. The same marginal pattern can arise from latent population structure that the analyst is not modeling. If the trial population is a mixture of treatment-responders and non-responders (in the latent-class sense developed at length in `analysis/report/03-latent-class-mixture-application/`), and class membership correlates with both BR strength and PB strength (responder-class participants have both higher BR and higher PB on average), then the marginal expected value of $BR \cdot PB$ over the unobserved class label has a non-zero covariance term whose magnitude depends on the gating function $\pi(B)$. From the one-component analysis perspective this looks exactly like a γ-weighted $BR \times PB$ interaction even though no such direct interaction was specified in the class-conditional DGP. That is, the latent-class structure is one mechanism that produces apparent subadditivity in a one-component analysis, regardless of whether the underlying biology has a true direct BR-PB interaction.

The two mechanisms, namely true subadditive interaction and latent-class structure with class-correlated components, produce qualitatively similar marginal effects but are inferentially distinct. Distinguishing them requires either a class-aware analysis that explicitly models the latent population structure (as the latent-class paper develops at length), or a phase contrast that separates the mechanisms by exploiting the differential expectancy modulation of PB that the BR-PB-TV framework provides. Under the additive-extension analysis fitted to mixture-DGP data with class-correlated components, $\hat{\gamma}$ is identified up to confounding with the class structure, and the analyst should report $\hat{\gamma}$ alongside a class-aware sensitivity analysis rather than as standalone evidence for mechanistic subadditivity. The Study 1 simulation in the latent-class paper includes a `pb_class_correlation` axis that varies $\mathrm{Cor}(BR, PB)$ across class labels; that axis is the direct simulation-side test of the latent-class generation of apparent subadditivity, and the cross-paper synthesis between the two papers is the cleanest way to disentangle the two mechanisms.

3.3 Time-variant natural-history component (TV)

TV is the reduction in symptom score that would have happened even if the patient had received no treatment at all. It is the participant's natural-history trajectory, integrated over the duration of the trial. For some participants TV is positive (symptoms naturally improve over the trial because of life events, therapy, time-in-condition, or regression to the mean); for others TV is negative (symptoms naturally worsen because of trauma anniversaries, accumulating stress, or progressive underlying disease).

Like PB, TV is not noise. It is a structured participant-specific signal that the model can in principle estimate from data, given sufficient repeated measurements and a trial design that includes a long enough off-drug baseline or follow-up window. What distinguishes TV from PB is its origin: TV is independent of treatment status and of belief about treatment, whereas PB is modulated by both.

3.3.1 When does natural history actually exist?

A common simplifying assumption in trial analysis is that subjects will improve over time, with or without treatment. This is true for some conditions and problematic for others. The shape of TV depends on the underlying disease class, the trial timescale, and the selection process by which participants entered the trial. None of these can be assumed without thought, and the empirical shape of TV is one of the things the decomposition is designed to recover.

Acute conditions tend toward TV-positive: a sprained ankle, a post-operative pain syndrome, an episode of acute back pain all resolve on a timescale of days to weeks even without intervention, and a trial of an analgesic in such a setting will see substantial TV-improvement in the placebo arm. Chronic stable conditions (treated hypertension, controlled chronic asthma, long-standing PTSD without active triggers) tend toward TV-zero on the population mean but with substantial individual variance. Chronic progressive conditions (Alzheimer's disease, ALS, late-stage congestive heart failure, many cancers) tend toward TV-negative: a trial of a

neuroprotective agent in early Alzheimer's must explicitly model the negative natural-history trajectory, since the apparent treatment effect is the slowing of progression, not absolute improvement. Cyclical and triggered conditions (PTSD with trauma anniversaries, seasonal-pattern major depression, relapsing-remitting multiple sclerosis) show structured TV that is neither monotone nor stationary, and may require explicit covariates beyond the Gompertz form.

Selection-induced TV is real and rarely zero, even when the underlying condition is stable. Patients enroll at moments of clinical engagement that often coincide with recent worsening, so regression to the mean alone produces several points of apparent TV-improvement in the early weeks of a trial. As the introduction documents, the systematic-review evidence of[13],[14], and[15] (the last covering 202 trials across 60 conditions) and the antidepressant analyses of[8] and[9] show that much of what is conventionally attributed to placebo response is indistinguishable from regression to the mean and natural-history drift.

Trial participation itself also induces TV: the Hawthorne effect, the structured-visit effect, and the diary-completion effect appear in the data as TV-improvement that is independent of any pharmacological or placebo response.

3.3.2 Can we assume no TV effect?

But how safe is it to assume TV is zero? The short answer is: rarely safe, and never safe without empirical support. The longer answer is that TV may be assumed zero only if the disease is in a stable chronic phase, the trial is short relative to the disease timescale, the recruitment process does not favor participants at peak severity, and the trial protocol does not introduce ancillary care that itself improves symptoms. For most chronic trials in psychiatric or pain conditions, at least one of these conditions fails. The conservative default is to estimate TV from the data, even at the cost of some statistical efficiency, rather than assume it is zero.

The decomposition makes this assumption testable. A model that includes a TV component will return $\hat{m}_{TV}$ with a variance estimate; if both are small relative to $\hat{m}_{BR}$ and $\hat{m}_{PB}$, the analyst has empirical support for the simplifying assumption. If $\hat{m}_{TV}$ is meaningfully large, the assumption was wrong, and the trial needs the TV component to produce unbiased inference about the others. Setting TV to zero and absorbing it into ε is worse than estimating it: the participant-level TV is structured (each participant has their own slow trajectory) and not independent of the on-drug-versus-off-drug schedule, so absorbing it into noise produces residuals that are autocorrelated within participants and heteroscedastic across them. The resulting standard errors on the BR and PB estimates are wrong in both directions: too small for participants whose TV is small, too large for participants whose TV is large. Estimating TV explicitly fixes both problems simultaneously.

3.3.3 Why a Gompertz, not a linear time covariate?

A natural objection is that a model with `week` as a covariate already controls for time. But how adequate is such a control? Three reasons suggest it is not enough. First, a `week` covariate enforces a linear trajectory, constraining each additional week to contribute a fixed amount of natural-history change; real disease trajectories curve, plateau, or accelerate, and the Gompertz form captures the S-shaped or saturating patterns linear time cannot. Second, a single `week` coefficient estimates a population-mean trajectory that ignores the participant-level heterogeneity in natural history. Third, BR and TV both evolve smoothly over the same trial timescale, and a `week` covariate cannot tell them apart from observational data alone; the design contrasts discussed in section 4 are what supply identifiability, and without the explicit decomposition the `week` coefficient absorbs both.

4 Identifiability and trial-design contrasts

The mathematical decomposition above is meaningless unless the data contain enough information to pin down each component separately. Trial design is what supplies that information. In this section we shall characterize the hybrid open-label-plus-blinded-discontinuation design, used as the worked example throughout this paper, and show how each of its three phases contributes a different combination of components to the observed trajectory.

Phase	Drug?	Belief?	Components present
Open-label on drug	Yes	Full	$BR + PB(\eta = 1) + TV$
Blinded discontinuation	Hidden	Hedged	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 0.5) + TV$
Open-label crossover	Either	Knows	$BR \cdot \mathbf{1}_{\text{on drug}} + PB(\eta = 1) + TV$

Three phase-by-allocation contrasts then identify the three components. The open-label versus blinded contrast (within drug) yields $PB(\eta = 1) - PB(\eta = 0.5) \approx 0.5 \cdot PB$, which isolates the belief component, but only conditional on a known blinded-phase expectancy η: from a two-phase design just the product $\eta \cdot m_{PB}$ is identified, and separating η from m_{PB} requires the open-label placebo phase. The blinded on-drug versus blinded placebo contrast yields BR once drug carryover has elapsed, since PB and TV match across allocation arms within the blinded phase. All phases off drug yield TV plus any residual off-drug PB, with no active BR contribution beyond carryover and PB at the off-drug expectation level. The $\mathbf{1}_{\text{on drug}}$ indicator in the phase table above is the binary drug state D_{it} of Table 1; the analysis carries the exposure-decayed $D_{bc,it}$ in its place.

This conditioning on η is the framework's principal identification limitation and deserves emphasis rather than a parenthetical. With a two-phase (open-label and blinded) design

the data identify only the product $\eta \cdot m_{PB}$, so PB is recovered up to the assumed blinded-phase expectancy; the working value $\eta \approx 0.5$ is a modeling assumption, not an estimate, and absent a belief or expectancy measurement it is not falsifiable from the outcome data alone. The identified set for m_{PB} is accordingly the interval $\{\widehat{\eta m_{PB}}/\eta : \eta \in (0, 1]\}$, which collapses to a point only when η is pinned down. Two design features close the gap: an open-label placebo phase, which supplies a second equation separating η from m_{PB}, and direct manipulation of belief, as in the open-hidden and balanced-placebo designs[53,54] that vary the patient's treatment expectation independently of exposure. In a balanced-placebo design what the patient is told and what the patient receives are crossed, so that drug exposure and treatment belief vary independently. We therefore recommend that any trial relying on the decomposition for a PB-versus-BR attribution either include such a phase or collect an expectancy measure, and that analyses report sensitivity of the attribution to η across its plausible range.

A trial that lacks any of these phases loses the corresponding identifiability. A pure open-label design (no blinded discontinuation) cannot separate BR from PB, because both are present at full strength throughout. A pure blinded design (no open-label phase) cannot estimate $\eta = 1$ PB, because no phase has belief at full strength. The decomposition machinery is useful only to the extent that the design supplies identifiability, and trial designers should evaluate any proposed design by checking that each component has at least one phase or contrast in which it is uniquely present.

Identifiability in the design-contrast sense above is necessary but not sufficient for stable estimation at finite sample sizes. The three Gompertz trajectories share an onset shape and overlap in calendar time, so their parameters can be collinear even when the contrasts that identify them in principle are present. Two conditions make the joint estimation credible: the design must supply at least one phase in which each component varies while the others are held fixed (the blinded-placebo contrast for BR, the open-label-versus-blinded contrast for PB, and an off-drug window for TV), and that off-drug window must be long enough relative to the carryover half-life for TV to separate from a still-decaying BR. We therefore do not assert that the three components are always recoverable. The full nonlinear component-matched fit that would map the recoverable region was not implemented for this paper (section 7.8), and the participant-level claims of the framework should be read as conditional on an identifiability that this paper does not demonstrate. A separate point, easy to confuse with collinearity, is that the phase-augmented formula of section 5.2 is exactly rank deficient on any design whose open-label visits are all on drug. That deficiency is a property of the design matrix at every N, recorded in the driver's deviations log on 2026-05-07, and section 5.2 describes the formula that was fitted in its place.

5 Analysis-side methodology

In this section we shall take up the practical question of what analysis model to use. A reasonable reaction to the formal model in section 2 is the following: if we are simulating data with three components plus noise, why is the analysis model written as `Sx ~ bm + t`

`+ Dbc + bm:Dbc` with a single random intercept and AR(1) residual correlation? Should we not include three parametric Gompertz components in the analysis, mirroring the DGP, so that the analyst sees the same structure that generated the data? The question is reasonable. Four constraints make a component-matched analysis impractical for most aggregated N-of-1 trial sizes, and a modest extension (a phase-by-treatment indicator) captures the practically important part of the gap at minimal cost.

5.1 Four reasons not to match the DGP

The first reason is identifiability. The trial design must supply the contrasts that identify each component, and many designs supply only part of it. The decomposition is identified by phase contrasts (open-label versus blinded for PB, on-drug versus blinded-placebo for BR, off-drug timepoints for TV). A pure open-label trial has none of these contrasts; in such a trial BR and PB both rise from zero on treatment initiation and are statistically indistinguishable from each other, regardless of how the analyst writes the model. Forcing an analysis to fit three components when the design supplies contrasts for only one or two of them does not recover the missing components; it produces unidentified or weakly identified parameter estimates with implausibly tight standard errors that misrepresent the analyst's actual inferential precision.

The second reason is degrees of freedom. Each component term costs degrees of freedom that come out of the moderation estimand. The biomarker-moderation question (the core scientific aim of the predictive-biomarker validation program) is asked through the `bm:Dbc` term. A component-matched analysis adds a Gompertz BR (three parameters), a Gompertz PB (three parameters plus a phase-specific expectancy multiplier), and a Gompertz TV (three parameters), each with participant-specific random effects on the maxima. That is on the order of a dozen new parameters, half of them on a non-linear scale. At trial sample sizes of $N \in [30, 150]$, the variance of $\hat{\beta}_{bm:D}$ is dominated by participant and timepoint counts, and adding parametric structure to absorb residual confounding usually inflates the moderation estimand's variance by more than it reduces bias.

The third reason is convergence fragility. Component-matched analyses are non-linear in the parameters and their convergence is problematic in practice. Fitting Gompertz BR, PB, and TV simultaneously requires a non-linear mixed model (`nlme::nlme` with self-starting Gompertz forms), a Bayesian hierarchical fit, or a structural-equation approximation. Each is more sensitive to starting values, more prone to local optima, and more likely to fail to converge on a single trial dataset than the linear-mixed analogue. We did not fit such a model in this paper and report no convergence rate for it.

The fourth reason is that, for a biomarker uncorrelated with PB and TV, the one-component `bm:Dbc` interaction already equals the pharmacological slope (section 6.1). In that case the regulator's question, whether the biomarker predicts treatment response under the trial conditions, and the pharmacological question coincide. They separate only when the biomarker is correlated with a non-pharmacological component, which is the case section 8 treats.

It is worth stating the inferential target of the analysis in the vocabulary of the estimands

framework[55], because the biomarker-validation question is easy to pose ambiguously. The primary estimand is the population-level biomarker-treatment interaction, the $bm{:}D_{bc}$ coefficient: its population is the trial-eligible patient population, its treatment contrast is on-drug versus off-drug exposure summarized through the continuous indicator D_{bc}, its endpoint is the symptom trajectory, its population-level summary is the interaction coefficient, and its intercurrent events are handled by a treatment-policy strategy for dropout and by the modeled exponential decay for residual carryover. The pharmacological-component target β_{bm}^{BR} is a more ambitious estimand: it is the biomarker-by-BR interaction that would obtain with the belief-driven channel held at its off-drug level, identified here through the phase contrasts of section 4 rather than by assumption. We flag explicitly that recovering a participant-level interaction of this kind inherits the difficulty Senn has emphasized for individual treatment effects[51,56]: a genuine participant-by-component signal is separable from participant-by-period and within-person random variation only when the design supplies repeated phase contrasts, and the framework should be read as estimating these quantities under that condition rather than as dissolving the well-known problem of identifying individual effects from finite within-person data.

5.2 Two linear-mixed extensions: phase indicators and expectancy terms

The smallest step toward component-aware analysis, without paying the costs above, is to add a phase indicator and its interactions with the drug indicator. The working protocol specified

```
 Sx ~ bm + t + Dbc + phase + Dbc:phase + bm:Dbc + bm:Dbc:phase
+ (1 | ptID), correlation = corCAR1(...)
```

with `phase` a two-level factor (open-label against all other visits, set by $\eta_t \geq 0.99$). The `Dbc:phase` term is not estimable on any design whose open-label visits are all on drug, which includes every design in this paper. There $D_{bc} = 1$ in the reference phase, so the `Dbc` column is an exact linear combination of the intercept, the `phase` column and the `Dbc:phase` column, at every N. The driver's deviations log recorded this on 2026-05-07 and `nlme::lme` stops on the full formula with a singularity error. The phase-augmented analysis reported in section 7 is therefore the formula without that term,

```
 Sx ~ bm + t + Dbc + phase + bm:Dbc + bm:Dbc:phase
```

and its `bm:Dbc` coefficient is an open-label versus other-phase contrast in which drug state and expectancy change together. The stored Study 1 replicates record `formula_dropped` as `Dbc:phase` for 99,995 of the 100,000 phase-augmented fits and `Dbc:phase+all` for the remaining 5; the June contamination replicates record it for every fit. The `bm:Dbc:phase` term tests whether the moderation shrinks when belief is hedged, a signature that the interaction was partly biomarker-by-PB. The analysis adds two columns, does not separately estimate BR and PB, and introduces a secondary family of belief-attenuation tests whose multiplicity should be acknowledged.

The second extension replaces the phase factor with the design's expectancy η_t itself, the option (b) of the deviations log:

```
Sx ~ bm + t + Dbc + De + bm:Dbc + bm:De
```

This is the F1 model of the project's analysis series, the specification the series recommends for the interaction and reports as robust to a prognostic biomarker on the standard Hybrid design (project documents `docs/47` to `docs/49` and `docs/56`). It is identified whenever D_{bc} and η_t take at least three distinct combinations across visits, which holds on every design here. Under the covariance encoding of the interaction, the biomarker-BR correlation is present on drug and loads on D_{bc}, while the biomarker-PB correlation scales with η_t and loads on η_t; the `bm:Dbc` coefficient of F1 therefore targets β_{bm}^{BR} and the `bm:De` coefficient absorbs the PB channel, to the extent that D_{bc} and η_t vary separately. Studies 3 and 4 fit this model under the name `decomposition` in the stored data; the text calls it F1. The series fits F1 with cluster-robust (CR2) standard errors; the fits here use model-based standard errors, since the estimand is the coefficient and not its test.

5.3 When to fit the full decomposition

A component-matched analysis merits the effort when sample size is in the several hundreds (a phase-3 trial fielded as aggregated N-of-1, or a meta-analysis pooling multiple smaller trials); the trial design provides timepoints in every phase combination, with enough density per cell to constrain the Gompertz curves; the scientific question requires the BR-PB split (a pharmacology paper claiming a specific mechanism, where ruling out a placebo contribution is the central point); and a Bayesian hierarchical fit with informative priors on the Gompertz parameters is feasible. In that regime, `lcmm`, `nlme::nlme` with self-starting Gompertz forms, or a `brms` hierarchical formulation can return participant-specific BR, PB, and TV component estimates that mirror the DGP. Outside that regime, the linear-mixed analysis with phase indicators is the better tool.

The asymmetry between DGP and analysis is not a flaw of the framework; it is a deliberate consequence of the asymmetry between simulation (where the analyst controls the DGP and can specify any structure, however elaborate) and inference (where the analyst must fit a model to a finite dataset and pay the identifiability cost of every additional parameter). Simulation studies use the rich DGP to characterize how a simpler analysis behaves under realistic data generation; they do not require the analysis to match the DGP in order to produce useful inference.

A word on the carryover parameterization is in order, since it is the one place where the analysis model makes a strong functional-form commitment. The continuous drug indicator D_{bc} decays as $\exp(-\lambda t_{sd})$ off drug, a one-parameter exponential governed by the carryover half-life. This is deliberately parsimonious and is best understood as a constrained special case of the more flexible carryover models now available for repeated-measures and N-of-1 data. Liao et al.[33] model carryover through a Bayesian distributed-lag structure with autocorrelated errors, relaxing the single-rate exponential assumption at the cost of additional

parameters; Daza[57] formalizes an average-period treatment-effect estimand built to accommodate autocorrelation, trend, and slow carryover jointly; and G{"a}rtner et al.[58] benchmark linear models against G-estimation and Bayesian-network alternatives specifically under carryover. We adopt the exponential form because it ties directly to a pharmacokinetic half-life and conserves degrees of freedom for the moderation estimand at trial-relevant sample sizes, but the choice is an assumption rather than a finding. A distributed-lag analysis arm that would quantify the cost of the restriction was not run (section 7.8).

6 Worked example: information loss without decomposition

6.1 What the one-component analysis estimates

Before the numerical illustration it is worth establishing algebraically what the one-component analysis actually estimates, because the qualitative failure modes of section 1 are consequences of an identity rather than empirical tendencies that require simulation to demonstrate. Write the within-participant symptom reduction as $R_{it} = \mathrm{BL}_i - Y_{it} = BR_{it} + PB_{it} + TV_{it} - \varepsilon_{it}$. The one-component analysis fits a single drug indicator $D_{bc,it}$. In symbols:

$$R_{it} \;=\; \alpha + \beta_D\, D_{bc,it} + g(t) + u_i + e_{it},$$

and reports $\hat{\beta}_D$ as 'the treatment effect'. Taking the on-drug minus off-drug contrast that β_D targets, and using that the natural-history component is by construction independent of treatment status so that its on/off difference cancels in a within-participant comparison, the probability limit of the estimator is:

$$\operatorname{plim} \hat{\beta}_D \;=\; m_{BR} + b_{PB} + b_{TV},$$

where $b_{PB} = E[PB \mid \text{on}] - E[PB \mid \text{off}]$ is the belief differential between the contrasted phases and b_{TV} collects any natural-history contribution that fails to cancel under the contrast actually used; b_{TV} is zero for a clean on/off within-participant contrast and strictly positive for a baseline-referenced pre-post analysis, in which case $b_{PB} = E[PB]$ and $b_{TV} = E[TV]$. To fix which case is in play below: the one-component analysis used in the worked example and the simulations of this paper is the design-contrast (drug-on/off) form, so $b_{TV} \approx 0$ there and the operative displacement is the belief term b_{PB}; the baseline-referenced pre-post form, which additionally carries $b_{TV} = E[TV]$, is the worse case retained only for comparison. Both terms are generally non-negative for an improving outcome, so the one-component estimate over-states the pharmacological component m_{BR}. The displacement is a property of the probability limit and is unchanged as $N \to \infty$: a larger one-component trial converges on the same biased value. This is the precise sense in which the placebo-attribution failure of section 1 is a problem of identification, not of precision.

The same algebra settles the biomarker-validation failure. The precision-medicine analysis regresses a participant-level response summary on a baseline biomarker B_i. If that summary

is the one-component per-participant effect, which estimates $m_{BR,i} + w_{PB}\, m_{PB,i} + w_{TV}\, m_{TV,i}$ for design-determined loadings $w_{PB}, w_{TV} \geq 0$, then by the linearity of covariance the fitted biomarker slope has probability limit:

$$\beta_{bm}^{\text{one}} \;=\; \frac{\text{Cov}(m_{BR} + w_{PB} m_{PB} + w_{TV} m_{TV},\, B)}{\text{Var}(B)} \;=\; \beta_{bm}^{BR} + w_{PB}\, \beta_{bm}^{PB} + w_{TV}\, \beta_{bm}^{TV},$$

where $\beta_{bm}^{C} = \text{Cov}(m_C, B)/\,\text{Var}(B)$ is the component-specific slope. A biomarker correlated with placebo responsiveness or with the natural-history trajectory, so that β_{bm}^{PB} or β_{bm}^{TV} is non-zero, therefore registers a non-zero one-component slope even when its pharmacological association β_{bm}^{BR} is exactly zero. This is the biomarker-validation failure of section 1, and it too is an identity under the additive decomposition rather than an empirical tendency: no sample size separates the one-component slope from a genuine pharmacological one, because the two coincide in the estimand whenever $w_{PB}\beta_{bm}^{PB} + w_{TV}\beta_{bm}^{TV} \neq 0$.

We make two qualifications explicit, because they bound how much weight the framework can carry. First, both displays are immediate corollaries of the additive decomposition and the linearity of the projection that defines the estimator; they are not a derived result in any deeper sense, and their value is in naming the displacement terms, not in the algebra. Second, they assume the components combine additively. Under the subadditive interaction documented in section 3.3, where the joint effect of BR and PB falls below their sum, a first-order correction proportional to $\text{Cov}(m_{BR} \cdot m_{PB}, B)$ enters the biomarker slope, and the identities above should be read as the leading behavior in the additive case rather than as exact under interaction. The empirical magnitude of that correction is not characterized here (section 7.8).

The loading w_{PB} can be written out for the data-generating process of section 7, and the result predicts each simulation before it is run. Under the covariance encoding of the interaction, $\text{Cov}(B, BR_{it}) = c_{bm}\, \sigma_{BR}$ on drug and decays off drug, while $\text{Cov}(B, PB_{it}) = c_{bm,PB}\, \sigma_{PB}\, \eta_t$ at every visit because the PB amplitude scales with expectancy. The one-component `bm:Dbc` coefficient is an on-minus-off within-participant contrast of the biomarker slope. Its probability limit is therefore

$$\beta_{bm}^{BR} \;-\; c_{bm,PB}\, \sigma_{PB}\, (\bar{\eta}_{\text{on}} - \bar{\eta}_{\text{off}}), \qquad w_{PB} = \bar{\eta}_{\text{on}} - \bar{\eta}_{\text{off}},$$

where the bars are the design's mean expectancy over on-drug and off-drug visits and the sign is negative because the components are reductions in Y. The sign of the displacement is a property of the design. It is negative when belief is higher on drug than off, as in the hybrid design, and positive when the reverse holds, as in the balanced-placebo design of Study 4. Its size does not involve m_{PB}. The approximation ignores the decay of D_{bc} at off-drug visits and the generalized-least-squares weighting of the mixed-model contrast, and section 7 reports how closely the simulated cells follow it.

Two consequences for how the framework should be evaluated follow. First, the quantity that carries the bias is the location of the estimand, not its sampling variability: the one-component coefficient remains a well-estimated number with a valid Wald test of its own

nullity, and that test rejects at the rate dictated by the displaced value. A test of nullity cannot distinguish a displaced coefficient from a true effect of the displaced size, whatever the direction of the displacement. Detecting the displacement requires comparing $\widehat{\beta}_D$ against m_{BR}, or the biomarker slope against β_{bm}^{BR}, which is exactly the comparison the decomposition, or a phase contrast that isolates the pharmacological part, makes available and the one-component analysis does not. Second, and as a corollary for the simulation evidence, a rejection-rate or power summary is the wrong instrument for this bias: the right target is the coefficient itself and its offset from the pharmacological value. This is why the pilot of section 7.2 found the biomarker-treatment rejection rate essentially flat across the placebo-strength axis even though that axis is precisely what drives the predicted bias; the prediction lives in the mean of $\widehat{\beta}_{bm:D_{bc}}$, and a run that means to test it must read the bias off the coefficient, not off the test.

6.2 A numerical illustration

To illustrate the inferential cost of failing to decompose, we shall work through a simple numerical example. Consider two participants, A and B, enrolled in a sixteen-week N-of-1 trial of prazosin for PTSD. Both are otherwise similar: same age, same baseline severity, same trial design. They differ in two latent traits that the trial does not measure directly. Participant A is a high placebo responder and a moderate pharmacological responder, with true component values $m_{BR}^{A} = 4$, $m_{PB}^{A} = 6$, $m_{TV}^{A} = 1$. Participant B is a low placebo responder and a strong pharmacological responder, with $m_{BR}^{B} = 8$, $m_{PB}^{B} = 1$, $m_{TV}^{B} = 1$.

In the open-label phase (weeks 1 to 8), both BR and PB run at full strength and TV accumulates. By week 8 both participants have reached close to their saturation values:

| BR | $PB(\eta = 1)$ | TV | Total improvement | |—|—|—|—|—| |

Participant A | 4.0 | 6.0 | 1.0 | **11.0** | | Participant B | 8.0 | 1.0 | 1.0 | **10.0** |

A clinician examining only the totals would conclude that the two participants are responding similarly: both improved by about ten points, both look like clinical successes. A simple t-test on the open-label change would estimate a population drug effect of about ten points and would not distinguish the two participants.

In the blinded discontinuation phase (weeks 9 to 12), both participants are silently switched to placebo at the start of week 9. The expectancy multiplier drops from 1.0 to 0.5. For clarity of exposition we take the carryover half-life to be short relative to the four-week blinded window, so that BR has effectively cleared by the end of week 12; the more realistic case of a slowly decaying BR, modeled by the carryover machinery of section 3.1, attenuates but does not eliminate the contrast below. By the end of week 12, the components are:

| BR | $PB(\eta = 0.5)$ | TV | Total improvement | |—|—|—|—|—|
Participant A | 0.0 | 3.0 | 1.0 | **4.0** | | Participant B | 0.0 | 0.5 |

1.0 | **1.5** |

The two participants now look quite different. Participant A still shows a sizeable improvement (four points) under blinded placebo; participant B has nearly returned to baseline (one

and a half points). This is the diagnostic contrast that the decomposition exploits. For simplicity the table holds TV at its week-8 value. With the calibrated TV parameters ($d_{TV} = 5$, $r_{TV} = 0.42$ per week) and $m_{TV} = 1$, TV rises from 0.74 at week 8 to 0.92 at week 12, an increment of 0.18 for both participants, which shifts both totals by a common amount and leaves the BR-versus-PB contrast on which the example turns unchanged.

A one-component fixed-effects regression of total symptom change on phase-by-allocation indicators sees:

- Open-label means: 11.0 (A), 10.0 (B). - Blinded-placebo means: 4.0 (A), 1.5 (B). - Estimated drug effect: $11.0 - 4.0 = 7.0$ (A), $10.0 - 1.5 = 8.5$ (B). Population mean: 7.75. - Estimated placebo effect: not identifiable from this contrast. - Estimated natural history: not identifiable from this contrast.

The one-component model has produced a single number conflated with the placebo response that washes out under blinding. It cannot say whether the drug effect is the same for participants A and B; it estimates 'drug' at 7.75 points on average, which is an average of the true 4.0 (A) and 8.0 (B) mixed with the placebo wash-out, with no principled way to decompose further.

A model with the full BR-PB-TV decomposition, fitted to the same data, can in principle recover the individual components by exploiting the differential expectancy between phases and the on-drug-versus-blinded-placebo contrast. In the noiseless limit, with the components exactly additive and η known, it returns $\hat{m}^A_{BR} = 4.0$, $\hat{m}^B_{BR} = 8.0$, $\hat{m}^A_{PB} = 6.0$, $\hat{m}^B_{PB} = 1.0$, $\hat{m}^A_{TV} = \hat{m}^B_{TV} = 1.0$. No such model is fitted in this paper, and participant-level estimates at trial-relevant N carry an uncertainty that this paper does not characterize (section 7.8).

The information loss from combining the components into one indicator is not abstract. We may compare the inferences directly:

Question | One-component answer | Three-component answer | |—|—|—| |

Does the drug work for A? | ~7.75 points (one-component). | $BR_A = 4.0$, distinct from her larger placebo response. | | Does the drug work for B? | ~7.75 points (one-component). | $BR_B = 8.0$, twice the magnitude of A. | | Are A and B different responders? | No discernible difference. | B has 2x A's BR but 1/6 of A's PB. | | Is a biomarker predicting BR useful here? | Cannot answer. | Yes if the biomarker correlates with $\hat{m}_{BR}$. | | Should A be continued long-term? | Cannot say. | Probably yes ($BR = 4$ is real and pharmacological), but the larger placebo component will not persist if expectations decay. | | Population mean drug effect? | 7.75 points (biased). | $6.0 = (4.0 + 8.0)/2$ (correct mean of BR). |

The one-component model has produced one biased average where the three-component model has produced six clinically actionable quantities. Predictive-biomarker validation, which is a central scientific aim of the trial design family considered here, becomes impossible without the decomposition.

A second example at the population level illustrates the same problem. Consider two cohorts of N-of-1 participants enrolled in methodologically identical trials of the same drug. Cohort

1 (early trial, naive patients) has mean $m_{BR} = 5$, mean $m_{PB} = 7$, mean $m_{TV} = 1$. Cohort 2 (later trial, patients who have had multiple prior failed treatments and are skeptical) has mean $m_{BR} = 5$, mean $m_{PB} = 2$, mean $m_{TV} = 1$. By construction the drug works equally well in both cohorts. But the open-label totals differ substantially: cohort 1 reports thirteen-point improvements on average, cohort 2 reports eight-point improvements. A one-component analysis comparing the two cohorts would conclude that the drug works less well in cohort 2 and might trigger a search for confounders or sub-population effects. None of this is real; the difference is entirely in PB, specifically in the population baseline level of placebo responsiveness. The three-component decomposition, applied to either cohort, returns $\hat{m}_{BR} = 5$ in both. Unfortunately, this is the kind of apparent non-replicability that pharmacology has spent the last decade struggling to navigate; the decomposition is one of the formal tools that makes it tractable.

7 Simulation studies

The simulation studies reported in this section are structured under the[59] ADEMP framework, specifying Aims, Data-generating mechanisms, Estimands, Methods, and Performance measures. The working ADEMP protocol is at `analysis/scripts/component-decomposition/00-ademp-protocol` and its deviations log at `02-deviations-log.md` in the same directory. The studies are exploratory: no analysis plan was registered in advance, and the findings should be confirmed in an independent study. Every study is calibrated to the prazosin-PTSD reference parameters ($m_{BR} = 10.99$, $\sigma_{BR} = 5$, $\sigma_{PB} = 2$, $\sigma_{TV} = 5$, $c_{bm} = 0.45$, $t_{1/2} = 1$ week, $\rho = 0.7$, $c_1 = 0.1$, $c_\times = 0.05$), with the interaction encoded as a correlation between the biomarker and the drug-response component that is present on drug and decays off drug. The pharmacological slope is $\beta_{bm}^{BR} = -c_{bm}\,\sigma_{BR}/\sigma_{bm} = -2.25$ in every alternative cell and 0 in every null cell. Bias is the mean estimated `bm:Dbc` coefficient minus that value, with the Monte Carlo standard error (MCSE) of the mean beside it. Table 2 lists every run the paper draws on. The protocol lettered its studies; following the project notation the studies are numbered here, and the June 2026 runs are identified by their stored directory names, which keep the old letters.

Table 2: Simulation runs reported in this paper. Replicates are given as alternative / null where the two differ. Directories are under analysis/data/derived_data/component-decomposition/ unless stated. Seeds enter as set.seed(seed + 1000 cell + replicate).

Study	Design	Cells	Replicates	Seed	Directory
Pilot	eight-visit	6	100	20260507	quick-sim/06-decomp-replicates.rds
Study 1	eight-visit ($N \leq 70$), sixteen-visit ($N \geq 200$)	80 alternative, 4 null	1,000 / 5,000	642	study-a-alt1000-null5000
Study 2	eight-visit ($N = 70$), sixteen-visit ($N = 300$)	16 alternative, 2 null	100 / 100	644	study-contam-alt100-null100
Study 3	sixteen-visit	8	1,000	815	study-b-recovery-pilot
Study 4	balanced-placebo	8	1,000	916	study-b-balanced
Corrected Study 2	eight-visit, sixteen-visit, balanced-placebo	22 alternative, 6 null	1,000 / 2,000	2061	study-2-feasible (not yet run)

7.1 Feasibility of the simulated cells

A correlation between the biomarker and two components at once is bounded by positive definiteness of the joint covariance matrix. With the biomarker row removed, the matrix R_0

does not depend on any biomarker correlation, and the biomarker row is $r = c_{bm}\, g + c_{bm,PB}\, p$ with g the drug-response pattern (1 on drug, decaying off drug) and p the placebo-response pattern (1 at every visit). The matrix is positive definite if and only if $r^\top R_0^{-1} r < 1$ (Schur complement), a quadratic in $c_{bm,PB}$ at fixed c_{bm} whose positive root is the largest feasible contamination. The project derived this bound for paper 02 after a strong prognostic correlation there turned out to be infeasible, and the drivers of that paper had repaired the matrix without warning (project document `docs/65`). The same happened here. The June 2026 drivers passed `makePositiveDefinite = TRUE` to the package, which at that time repaired a non-positive-definite matrix with `corpcor::make.positive.definite` and emitted no warning. The package has refused such matrices by default since 2026-10-26, and the repair is now locked behind an option that no driver of this paper sets.

Table 3 gives the bound for each design at $c_{bm} = 0.45$ and, for every contamination level used in any run, the smallest eigenvalue of the raw matrix, computed by `05-feasibility-bounds.R`. The bound is 0.321 on the eight-visit design, 0.315 on the balanced-placebo design and 0.126 on the sixteen-visit design, whose longer on-drug run and shorter visit spacing leave less room in the biomarker row. The eigenvalue bounds were verified by direct evaluation at 0.999 and 1.001 times each bound. Three consequences follow. Every Study 1 cell is feasible, since Study 1 sets $c_{bm,PB} = 0$, and feasibility does not depend on m_{PB} or m_{TV}, which enter the means only. Every Study 4 cell is feasible, since its grid stops at 0.30. In Studies 2 and 3, the sixteen-visit cells at $c_{bm,PB} \geq 0.15$ and the eight-visit cells at 0.45 drew from a repaired matrix. For those cells the table also gives what the repair did: the on-drug biomarker-BR correlation it produced, the pharmacological slope that the repaired process implies, and the population value of the one-component contrast under the repaired process. The repair lowers the on-drug correlation from 0.45 to between 0.45 and 0.36, and it also compresses the variances, so the implied pharmacological slope moves from -2.25 to between -2.21 and -1.44 on the sixteen-visit design. No cell reported below as a contamination result drew from a repaired matrix. Cells that did are reported separately, in section 7.4 and 7.5, as exhibits of the artifact, with bias measured against the repaired population value.

Table 3: Feasibility of the contamination cells at $c_{bm} = 0.45$. The bound is the largest $c_{bm,PB}$ for which the joint covariance matrix is positive definite, minimum over the design's paths. A negative smallest eigenvalue marks a cell that the June 2026 runs drew from a repaired matrix; the last three numeric columns give the on-drug biomarker-BR correlation after the repair, the pharmacological slope the repaired process implies, and the population value of the one-component contrast under the repaired process (the sum of that slope and the expectancy displacement of section 6.1). Nominal values are 0.45, −2.25 and −2.25 plus the displacement.

Design	Bound on $c_{bm,PB}$	$c_{bm,PB}$	Smallest eigenvalue	Repaired Cor(B, BR_{on})	Repaired β_{bm}^{BR}	Repaired one-component value
eight-visit	0.321	0.00	0.344			
eight-visit	0.321	0.05	0.343			
eight-visit	0.321	0.10	0.321			
eight-visit	0.321	0.15	0.277			
eight-visit	0.321	0.20	0.215			
eight-visit	0.321	0.30	0.044			
eight-visit	0.321	0.45	-0.307	0.399	-1.78	-2.10
sixteen-visit	0.126	0.00	0.045			
sixteen-visit	0.126	0.05	0.042			
sixteen-visit	0.126	0.10	0.020			
sixteen-visit	0.126	0.15	-0.023	0.446	-2.21	-2.33
sixteen-visit	0.126	0.20	-0.084	0.434	-2.09	-2.25
sixteen-visit	0.126	0.30	-0.260	0.404	-1.82	-2.03
sixteen-visit	0.126	0.45	-0.635	0.358	-1.44	-1.69
balanced-placebo	0.315	0.00	0.344			
balanced-placebo	0.315	0.05	0.339			
balanced-placebo	0.315	0.10	0.315			
balanced-placebo	0.315	0.15	0.271			
balanced-placebo	0.315	0.20	0.208			
balanced-placebo	0.315	0.30	0.032			
balanced-placebo	0.315	0.45	-0.348	0.390	-1.70	-1.25

7.2 Pilot validation of the analysis machinery

A 100-replicate-per-cell pilot was run on the eight-visit design at $N = 35$ over the six cells $m_{PB} \in \{0, 6.5, 13\}$ by analysis ∈ {one-component, phase-augmented}, with 8-core `parallel::mclapply`. Convergence was 1.0 in every cell. Per-replicate output is at `analysis/data/quick-sim/06-decomp-replicates.rds` and the cell-level summary at `06-decomp-summary.txt`. The one-component `bm:Dbc` test showed power 0.95, 0.93 and 0.83 across m_{PB}, the phase-augmented analysis 0.65, 0.65 and 0.61, and the mean estimate was flat within each analysis (about −2.25 and −2.21). At 100 replicates the MCSE of a power near 0.7 is about 0.046. The pilot established two things used below: the phase-augmented formula is rank deficient on this design (section 5.2), and the bias the identity predicts appears in the coefficient, not in the rejection rate. No substantive claim is drawn from it.

7.3 Study 1: bias and power with a biomarker correlated with the drug response alone

Study 1 ran 1,000 replicates per alternative cell and 5,000 per null cell over 80 alternative cells crossing $m_{PB} \in \{0, 1, 3, 6, 10\}$, $m_{TV} \in \{-1, 0, 1, 2\}$ and $N \in \{35, 70, 200, 300\}$, plus four null cells. The biomarker is correlated with BR alone, so the identity of section 6.1 predicts no bias, and none appears. Under the one-component analysis the mean absolute bias is 0.016 at

$N = 35$, 0.016 at $N = 70$, 0.006 at $N = 200$ and 0.006 at $N = 300$, in every case within one MCSE (about 0.027, 0.019, 0.009 and 0.007) of zero, with no monotone trend across m_{PB} or m_{TV}. Study 1 varies the magnitudes of the unmodeled components, which the identity shows to be the innocuous axis; the axis that produces biomarker-validation bias is the biomarker's correlation with PB or TV, which this study holds at zero.

Two features of the sample-size axis should be read with the results. N is the total across randomization paths. The two smaller cells use the eight-visit single-path design and the two larger cells the sixteen-visit two-path design with the crossover phase, so the step from $N = 70$ to $N = 200$ changes the design as well as the sample. Comparisons within either pair are interpretable; a comparison across the step is not attributable to sample size alone. The sixteen-visit design is also close to its feasibility bound even without contamination: its smallest eigenvalue at the reference parameters is 0.045 against 0.344 for the eight-visit design (Table 3), so any sensitivity analysis on it that raises a correlation must be checked against the bound first. A clean sample-size column on the eight-visit design at $N = 200$ and 300 was not run (section 7.8).

The comparison between analyses is clear. The phase-augmented analysis, fitted as the formula without the aliased `Dbc:phase` term in every replicate (section 5.2), is not less biased than the one-component analysis at any sample size (mean absolute bias 0.028 against 0.016 at $N = 35$, comparable elsewhere), and it is markedly less powerful for the biomarker-treatment test: mean power across the grid is 0.35 against 0.59 at $N = 35$ and 0.62 against 0.90 at $N = 70$, with both analyses at essentially full power by $N = 200$. Because the DGP contains no biomarker-PB correlation for the phase contrast to remove, the extra terms add estimation cost without bias reduction. Convergence was essentially complete for both analyses (minimum rate 0.998). The inverted power ranking is therefore a property of the fitted formula, whose `bm:Dbc` coefficient is an open-label versus blinded contrast; whether it would persist under an identified phase-augmented specification was not tested.

Calibration is adequate for both analyses. Nominal-95% confidence-interval coverage of $\beta_{bm:D_{bc}}$ ranges from 0.96 to 0.99 for the one-component analysis (slightly conservative) and from 0.94 to 0.99 for the phase-augmented analysis. Type I error at the four null cells is controlled, ranging from 0.031 to 0.056 for the one-component analysis and 0.038 to 0.049 for the phase-augmented analysis; the one-component estimate at $N = 200$ (0.056, MCSE 0.003) sits 1.8 standard errors above nominal.

Two conclusions follow. First, the biomarker-validation failure is real as an identity, but it is driven by the biomarker's correlation with PB or TV, not by the magnitude of those components. The decisive axis is $c_{bm,PB}$, which Studies 2 to 4 vary directly. Second, the phase-augmented analysis carries a power cost without a compensating bias reduction under an uncontaminated biomarker, and should not be a default for this design.

7.4 Study 2: a biomarker correlated with the placebo response

Study 2 (stored as `study-contam-alt100-null100`, 100 replicates per cell) varied $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$ at $m_{PB} \in \{0, 6\}$ on the eight-visit design at $N = 70$ and on the sixteen-

visit design at $N = 300$, fitting the one-component and the phase-augmented analyses. Before the results, the prediction. Table 4 gives each design's mean expectancy on and off drug and the loading $w_{PB} = \bar{\eta}_{\text{on}} - \bar{\eta}_{\text{off}}$. On the two hybrid designs belief is higher on drug than off, so the displacement $-c_{bm,PB}\,\sigma_{PB}\,w_{PB}$ is negative, about $-0.83\,c_{bm,PB}$ on the eight-visit design and $-0.80\,c_{bm,PB}$ on the sixteen-visit design. On the balanced-placebo design of Study 4 belief is lower on drug than off ($\bar{\eta}_{\text{on}} = 1/3$ against 1), so the displacement is positive, $+1.33\,c_{bm,PB}$. Its size does not involve m_{PB}.

Table 4: Expectancy loading of each design and the predicted displacement of the one-component biomarker slope, $-c_{bm,PB}\,\sigma_{PB}\,w_{PB}$ with $\sigma_{PB} = 2$. Expectancies are averaged over on-drug and off-drug visits with the binary drug state, pooled over paths.

Design	$\bar{\eta}_{\text{on}}$	$\bar{\eta}_{\text{off}}$	w_{PB}	Displacement at $c_{bm,PB} = 0.10$	at 0.20	at 0.30
eight-visit	0.917	0.500	+0.417	-0.083	-0.167	-0.250
sixteen-visit	0.900	0.500	+0.400	-0.080	-0.160	-0.240
balanced-placebo	0.333	1.000	-0.667	+0.133	+0.267	+0.400

Table 5 reports the feasible eight-visit cells beside the prediction. At $c_{bm,PB} = 0.15$ the one-component bias is -0.142 and -0.122 at the two m_{PB} levels (MCSE about 0.06) against a predicted -0.125. At 0.30 it is -0.374 and -0.432 against a predicted -0.250. The sign is the predicted one, the bias at 0.30 is six to seven MCSE from zero, and the two m_{PB} levels agree within one MCSE, as the identity requires. The simulated displacement at 0.30 exceeds the approximation by about 0.15, two to three MCSE at this replicate count; the approximation ignores the off-drug decay of D_{bc} and the generalized-least-squares weighting of the mixed-model contrast, and the corrected run of section 7.7 will measure the gap at 1,000 replicates. These cells were the only faithfully generated contaminated cells of the run, and an earlier version of this paper called them uninformative while building its narrative on the sixteen-visit cells; that reading is withdrawn. The phase-augmented analysis, fitted without its aliased term, shows the same displacement within its larger MCSE and offers no protection, as expected: both analyses see the biomarker-PB covariance through the same open-label versus blinded contrast.

Table 5: Study 2, feasible cells: eight-visit design, $N = 70$, 100 replicates per cell. Bias of the bm:Dbc coefficient against the pharmacological slope -2.25, with the displacement the identity predicts.

$c_{bm,PB}$	m_{PB}	Predicted displacement	One-component bias (MCSE)	Phase-augmented bias (MCSE)
0.00	0	-0.000	+0.062 (0.060)	+0.067 (0.095)
0.00	6	-0.000	-0.111 (0.063)	-0.162 (0.097)
0.15	0	-0.125	-0.142 (0.061)	-0.029 (0.092)
0.15	6	-0.125	-0.122 (0.065)	-0.108 (0.102)
0.30	0	-0.250	-0.374 (0.057)	-0.318 (0.093)
0.30	6	-0.250	-0.432 (0.061)	-0.352 (0.103)

The remaining cells of the run, at $c_{bm,PB} \geq 0.15$ on the sixteen-visit design and at 0.45 on

the eight-visit design, drew from repaired matrices. Table 7 reports them as an exhibit of the artifact together with the corresponding cells of Study 3. Against the nominal slope of -2.25 the sixteen-visit one-component mean at $c_{bm,PB} = 0.45$ is -1.74, an apparent bias of $+0.51$ some 30 MCSE from zero, which an earlier version of this paper reported as a contamination result. Against the repaired population value of -1.69 the bias is -0.05. The apparent positive, monotone bias was the gap between the nominal and the repaired data-generating process, not a property of the estimator, and the earlier statement that the contamination bias is toward zero, and its explanation through the biomarker main effect, are withdrawn. Under the repaired process the one-component estimates sit 0.03 to 0.07 below the repaired value in every exhibit cell, the same direction and order as the gap in the feasible cells.

7.5 Study 3: recovery on the sixteen-visit hybrid design

Study 3 (stored as `study-b-recovery-pilot`, 1,000 replicates per cell) fitted four analyses on the sixteen-visit design at $N \in \{140, 300\}$ and $c_{bm,PB} \in \{0, 0.15, 0.30, 0.45\}$: the one-component and phase-augmented analyses of Study 2, the F1 model of section 5.2, and a blinded-stratum analysis that fits `Sx ~ bm + t + Dbc + bm:Dbc` to the visits with $\eta_t < 1$ alone, where belief is constant and the drug-on/off contrast is uncontaminated by construction. No Gompertz-basis component decomposition was fitted, and the earlier statement that one was is withdrawn. Only the $c_{bm,PB} = 0$ cells are feasible on this design (Table 3); Table 6 reports them. All four analyses are unbiased there, within 0.02 of the pharmacological slope at MCSE 0.010 to 0.015, and F1 in particular estimates β_{bm}^{BR} without bias on a hybrid design whose drug-expectancy correlation is $\mathrm{Cor}(D_{bc}, \eta_t) = 0.753$, higher than the 0.485 of the standard four-path Hybrid design of the series. The contaminated cells are exhibits (Table 7). In them F1 and the blinded-stratum analysis return the pharmacological slope of the repaired process within 0.03, while the one-component and phase-augmented estimates carry the expectancy displacement. The earlier statement that no decomposition recovers the slope on the hybrid design rested on these cells measured against -2.25, and is withdrawn. Whether F1 recovers the slope at the contamination levels this design can accommodate, $c_{bm,PB} \leq 0.10$, is a question for the corrected run (section 7.7); the exhibits are consistent with recovery but were generated under a process that is not the one specified.

Table 6: Study 3, feasible cells: sixteen-visit hybrid design, $c_{bm,PB} = 0$, 1,000 replicates per cell. Bias of the bm:Dbc coefficient against −2.25.

N	Analysis	Bias (MCSE)	Power	Convergence
140	one-component	+0.006 (0.010)	1.000	1.000
140	phase-augmented	+0.014 (0.015)	0.992	1.000
140	F1	+0.018 (0.015)	0.989	1.000
140	blinded stratum	+0.018 (0.015)	0.992	1.000
300	one-component	-0.008 (0.007)	1.000	1.000
300	phase-augmented	-0.004 (0.010)	1.000	1.000
300	F1	+0.002 (0.010)	1.000	1.000
300	blinded stratum	+0.003 (0.011)	1.000	1.000

Table 7: Exhibit of the repair artifact: every contaminated cell of Studies 2 and 3 that drew from a repaired covariance matrix. The nominal value is −2.25 plus the identity displacement; the repaired value is the population value of the one-component contrast under the repaired process (the feasibility table). The last two columns compare the F1 estimate of Study 3 with the pharmacological slope the repaired process implies. None of these cells is a result about the specified data-generating process.

Study	Design	N	$c_{bm,PB}$	m_{PB}	One-component mean	Bias against −2.25	Nominal value	Repaired value	Bias against repaired	F1 mean	Repaired β_{bm}^{BR}
Study 2	sixteen-visit	300	0.15	0	-2.36	-0.11	-2.37	-2.33	-0.03		-2.21
Study 2	sixteen-visit	300	0.15	6	-2.37	-0.12	-2.37	-2.33	-0.04		-2.21
Study 2	sixteen-visit	300	0.30	0	-2.08	+0.17	-2.49	-2.03	-0.06		-1.82
Study 2	sixteen-visit	300	0.30	6	-2.08	+0.17	-2.49	-2.03	-0.05		-1.82
Study 2	sixteen-visit	300	0.45	0	-1.74	+0.51	-2.61	-1.69	-0.06		-1.44
Study 2	sixteen-visit	300	0.45	6	-1.74	+0.51	-2.61	-1.69	-0.05		-1.44
Study 2	eight-visit	70	0.45	0	-2.23	+0.02	-2.62	-2.10	-0.13		-1.78
Study 2	eight-visit	70	0.45	6	-2.31	-0.06	-2.62	-2.10	-0.21		-1.78
Study 3	sixteen-visit	140	0.15	0	-2.37	-0.12	-2.37	-2.33	-0.04	-2.22	-2.21
Study 3	sixteen-visit	140	0.30	0	-2.09	+0.16	-2.49	-2.03	-0.07	-1.80	-1.82
Study 3	sixteen-visit	140	0.45	0	-1.75	+0.50	-2.61	-1.69	-0.07	-1.43	-1.44
Study 3	sixteen-visit	300	0.15	0	-2.37	-0.12	-2.37	-2.33	-0.04	-2.20	-2.21
Study 3	sixteen-visit	300	0.30	0	-2.08	+0.17	-2.49	-2.03	-0.05	-1.79	-1.82
Study 3	sixteen-visit	300	0.45	0	-1.74	+0.51	-2.61	-1.69	-0.05	-1.43	-1.44

7.6 Study 4: recovery on a balanced-placebo design

Study 4 asks whether the displacement can be removed by the F1 model on a design that varies drug exposure and belief separately. The mechanism is that of section 5.2: the biomarker-BR correlation loads on D_{bc} and the biomarker-PB correlation on η_t, so interacting the biomarker with both separates the channels to the extent that the two covariates vary independently. The balanced-placebo design[53,54] used here has three open-label on-drug visits ($\eta = 1$), six covert on-drug visits in which active drug continues without the participant's knowledge ($\eta = 0$), and six concealed-placebo visits in which placebo is substituted while the participant is told treatment continues ($\eta = 1$), at 1.5-week spacing. The covert phase supplies drug exposure at zero belief and the concealed-placebo phase belief without drug, and $\mathrm{Cor}(D_{bc}, \eta_t) = -0.656$ against +0.753 on the sixteen-visit hybrid design. The design's feasibility bound is 0.315, so the grid $c_{bm,PB} \in \{0, 0.1, 0.2, 0.3\}$ lies inside it; at 0.45 the smallest eigenvalue is −0.35 and the cell was excluded. The covert visits have exactly zero PB variance, which is a degenerate component and not a feasibility failure; the June driver's repair call nudged those variances

to 10^{-3} and changed no correlation (largest change 0 at every grid value), and the package now factors the positive-variance block directly. The run (stored as `study-b-balanced`, seed 916) used 1,000 replicates per cell at $N \in \{70, 150\}$ and fitted the one-component analysis (stored as `lumped`) and F1 (stored as `decomposition`).

Table 8: Study 4: balanced-placebo design, 1,000 replicates per cell. Bias of the bm:Dbc coefficient against −2.25 for the one-component analysis and F1, with the identity's predicted displacement of the one-component slope and nominal-95% interval coverage.

N	$c_{bm,PB}$	Predicted displacement	One-component bias (MCSE)	One-component coverage	F1 bias (MCSE)	F1 coverage	F1 power
70	0.00	+0.000	-0.016 (0.024)	0.96	-0.034 (0.030)	0.94	0.66
70	0.10	+0.133	+0.166 (0.024)	0.94	-0.001 (0.031)	0.94	0.67
70	0.20	+0.267	+0.278 (0.024)	0.94	-0.076 (0.029)	0.96	0.69
70	0.30	+0.400	+0.453 (0.024)	0.93	-0.020 (0.031)	0.94	0.67
150	0.00	+0.000	-0.014 (0.017)	0.96	-0.022 (0.021)	0.94	0.93
150	0.10	+0.133	+0.161 (0.016)	0.94	+0.008 (0.020)	0.95	0.93
150	0.20	+0.267	+0.340 (0.016)	0.91	+0.021 (0.021)	0.95	0.93
150	0.30	+0.400	+0.476 (0.016)	0.86	-0.026 (0.020)	0.95	0.93

Table 8 gives the results. The one-component slope is displaced in the predicted direction, upward, by +0.161, +0.340 and +0.476 at $c_{bm,PB} = 0.1$, 0.2, 0.3 and $N = 150$ (MCSE 0.016), against predicted +0.133, +0.267 and +0.400. The $N = 70$ cells give the same displacement within MCSE. As on the eight-visit design, the simulated displacement exceeds the approximation, here by 15 to 25 percent. Its coverage falls to 0.86 at $c_{bm,PB} = 0.3$ and $N = 150$. F1 tracks the pharmacological slope across the whole grid: its bias at $N = 150$ is −0.022, +0.008, +0.021 and −0.026 (MCSE about 0.02), each within 1.4 MCSE of zero, with coverage from 0.94 to 0.96. At $N = 70$ three of the four F1 biases are within one MCSE of zero; the cell at $c_{bm,PB} = 0.2$ is −0.076 at MCSE 0.029, 2.6 MCSE from zero, with no trend across the grid, and we read it as Monte Carlo variation to be checked in the corrected run. F1's power is lower, about 0.67 at $N = 70$ against 0.93 at $N = 150$, reflecting the smaller sample and the extra terms. Convergence was complete in every cell.

The recovery is conditional on the design, and the designs that supply it require concealment. The balanced-placebo design of[54] deceives participants in two of its four arms, and the open-hidden paradigm of[53] administers drug without the participant's knowledge, which is feasible with concealed infusions in an inpatient setting. Translating either to a multi-week outpatient N-of-1 trial of an oral drug requires a consent procedure in which participants authorize being misinformed about allocation within the trial, the authorized deception discussed by[60], and the same authors note that disclosure adequate to protect participants may itself alter the expectancy the design means to manipulate. That such a design can be fielded in a psychiatric outpatient population is shown by the open-hidden antidepressant discontinuation protocol of[61] (NCT05191277), in which participants in the hidden arms are told there is a 50 percent chance of discontinuation. In the notation of this paper that instruction induces $\eta = 0.5$, the blinded-discontinuation value, rather than the $\eta = 0$ of a covert phase; a trial of that form supplies the D_{bc}-η_t decoupling through the crossing of continuation with instruction rather than through a zero-belief phase, and the F1 model applies to it unchanged. The recommendation that follows is specific: a biomarker-validation trial in which the candidate biomarker may correlate with placebo responsiveness should be fielded with an open-hidden or

balanced-placebo component, with the concealment authorized at consent, because only such a component makes the pharmacological biomarker-treatment interaction separately identifiable. Two questions remain open: recovery beyond the feasible contamination range cannot be studied under the correlation encoding, and whether the same recovery holds under the mean-shift encoding of the interaction, where the biomarker shifts component means rather than correlating with component draws, was not tested.

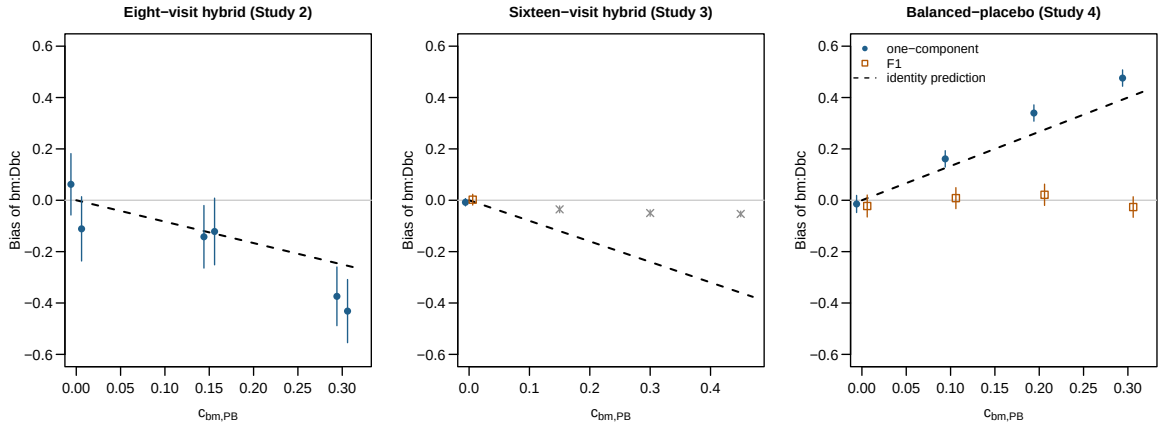


Figure 1: Bias of the one-component (filled circles) and F1 (open squares) estimates of the biomarker-treatment interaction against the biomarker-placebo correlation, one panel per design, with the displacement the identity predicts for the one-component slope drawn as a dashed line. Eight-visit: Study 2 feasible cells at $N = 70$, 100 replicates, two m_{PB} levels. Sixteen-visit: Study 3 at $N = 300$, 1,000 replicates; only the $c_{bm,PB} = 0$ cell is feasible, and the exhibit cells are shown as gray crosses with bias measured against the repaired population value. Balanced-placebo: Study 4 at $N = 150$, 1,000 replicates. Error bars are two MCSE.

Figure 1 collects the feasible cells of Studies 2 to 4 with the identity's prediction. The sign of the one-component displacement follows the design, its size follows the prediction to within about 20 percent, and F1 removes it where the design permits.

7.7 Corrected contamination study (results pending)

The driver `run-study-2-feasible.R` reruns the contamination study on feasible cells only, under the package's refusal of non-positive- definite matrices, with an independent seed (2061) and a pre-flight eigenvalue check of every cell. Its grid is the eight-visit design at $N \in \{35, 70\}$ and $c_{bm,PB} \in \{0, 0.10, 0.20, 0.30\}$, the sixteen-visit design at $N \in \{140, 300\}$ and $c_{bm,PB} \in \{0, 0.05, 0.10\}$, and the balanced-placebo design at $N \in \{70, 150\}$ and $c_{bm,PB} \in \{0, 0.10, 0.20, 0.30\}$, 22 alternative cells at 1,000 replicates, plus six null cells at 2,000 replicates that set $c_{bm} = 0$ with and without the biomarker-PB correlation, to check whether the expectancy-scaled placebo channel inflates the size of the interaction test. It fits the one-component analysis, the phase-augmented analysis (the identified formula, fitted directly) and F1, saves every replicate per cell, and reports bias, power and coverage with MCSE. A three- replicate smoke test on 2026-10-28 ran all 28 cells with no refused draw. The full run is estimated at about 45 core-hours from the smoke-test timing, roughly 7 hours on 7 local

cores or 12 to 15 hours on the simulation host, and has not been submitted. The command, from the repository root, is

```
Rscript analysis/scripts/component-decomposition/run-study-2-feasible.R \
  1000 2000 study-2-feasible
```

udy-2-feasible \end{verbatim}

RESULTS PENDING. The corrected contamination study has not been run. When analysis/data/derived_data/component-decomposition/study-2-feasible/study-2-summary.rds exists, this chunk renders the full results table in its place.

7.8 What was specified but not run

The following items appear in the working protocol or in earlier versions of this paper and were not run or implemented. They are listed so that no reader takes them as reported.

- The full nonlinear component-matched fit (protocol Study A method 3 and the whole of protocol Study B, the identifiability study). The driver's `full_nonlinear_fit()` is a stub that returns missing values. No convergence rate, variance-inflation factor or participant-level recovery measure exists.
- Protocol Study C, the subadditivity sensitivity study over γ and the latent-class comparison (section 3.3.1). Its driver `04-study-C.R` has no output.
- The η-mismatch sensitivity grid ($\eta_{\text{blinded}} \in \{0.25, 0.5, 0.75\}$), the TV-magnitude and dropout sensitivity grids, and a prognostic $c_{bm,TV}$ cell that would test the TV half of the identity.
- Study 1 at $N = 200$ and 300 on the eight-visit design, which would give a sample-size column unconfounded with the design change.
- A distributed-lag analysis arm (section 5.3).
- The production-resolution contamination run of the 2026-06 plan (`study-contam-alt1000-null5000`, one cell stored). It is superseded by the corrected study of section 7.7, which has not been run either.

8 Application to predictive-biomarker validation

The motivating clinical question for the trial designs in this project is whether a baseline biomarker predicts treatment response, and specifically whether the biomarker predicts the pharmacological component of treatment response. This is the precision-medicine question, formalized in the PATH statement[16]. Whether it depends on the BR-PB-TV decomposition is itself conditional: by the identity of section 6.1, the one-component biomarker slope equals the pharmacological slope when the biomarker is uncorrelated with PB and TV, and departs from it only by the biomarker's covariance with those components. The decomposition is

therefore the mechanism the question requires precisely in the regime where that covariance is non-negligible.

A direct regression of total response on baseline biomarker,

```
total_response = beta_0 + beta_bm * biomarker + error
```

estimates the biomarker's correlation with whatever the one-component total contains. If PB varies systematically across the biomarker distribution, because the biomarker is correlated with optimism, suggestibility, or treatment-seeking history, then $\hat{\beta}_{bm}$ contains a placebo-prediction component inseparable from the pharmacological-prediction component. The resulting biomarker, applied prospectively, would mis-stratify patients: those flagged as 'high responders' would include both true high-BR responders and high-PB responders, and the latter would not benefit from prescription decisions made on a pharmacological basis.

The right approach is to regress each component on the biomarker:

$$
\begin{aligned}
BR &= \beta_0^{BR} + \beta_{bm}^{BR} \cdot \text{biomarker} + \varepsilon^{BR}, \\
PB &= \beta_0^{PB} + \beta_{bm}^{PB} \cdot \text{biomarker} + \varepsilon^{PB}, \\
TV &= \beta_0^{TV} + \beta_{bm}^{TV} \cdot \text{biomarker} + \varepsilon^{TV}.
\end{aligned}
$$

A biomarker is a useful pharmacological predictor if and only if $\hat{\beta}_{bm}^{BR}$ is meaningfully non-zero. Significant $\hat{\beta}_{bm}^{PB}$ or $\hat{\beta}_{bm}^{TV}$ are diagnostic findings about the biomarker, indicating that it is partly psychological or correlates with natural-history characteristics, but they are not the validation of pharmacological prediction. A biomarker that correlates only with PB would be valuable for selecting placebo responders in run-in designs but worthless for predicting pharmacological efficacy; a biomarker that correlates only with TV would be a prognostic marker, not a predictive one.

The decomposition is therefore the mechanism by which the precision-medicine question is posed whenever the candidate biomarker may correlate with PB or TV. Where the biomarker can be assumed orthogonal to the non-pharmacological components, the identity of section 6.1 shows the one-component analysis already targets the biomarker-by-BR estimand, and Study 1 confirms it recovers that estimand without bias. Where the biomarker may be correlated with PB, the F1 model of section 5.2 targets β_{bm}^{BR} on a design in which drug exposure and belief vary separately (Study 4). The phase-augmented analysis, as fitted without its aliased term, reduced no bias under an uncontaminated biomarker and lost power (Study 1), and did not remove the contamination displacement on the hybrid designs (Studies 2 and 3). A full component-matched fit would recover the component-specific regressions explicitly; it was not fitted here.

9 Discussion

The three-component decomposition is the formal expression of the claim that 'treatment response' in an aggregated N-of-1 trial is not a single quantity but a sum of three causally distinct contributions, each with its own pharmacological, psychological, or epidemiological interpretation. What the present work establishes, analytically in section 6.1 and empirically in section 7, is that the inferential value of acting on this decomposition is conditional rather than universal. The identity shows that the one-component estimator is displaced from its pharmacological target only by the components' loading onto the contrast used, and, for a biomarker slope, only by the biomarker's covariance with PB and TV. Where those terms are zero, the design-contrast one-component analysis is already unbiased, and Study 1 confirms it: with a biomarker correlated with BR alone, the one-component biomarker-treatment estimate is unbiased across the whole PB-TV grid and the phase-augmented analysis is dominated. Where the biomarker is correlated with PB, the displacement follows the design's expectancy schedule, and the F1 model removes it on a design that varies exposure and belief separately. The framework's contribution is therefore best read as a map of when decomposition changes inference, not as a blanket prescription to decompose. It makes the trial-design conditions explicit, identifies the attribution and contaminated-biomarker estimands for which decomposition is needed, and clarifies the target estimand for biomarker validation (the biomarker-by-BR association, not the one-component total).

Four open questions remain and warrant acknowledgement. First, the corrected contamination study of section 7.7 has not been run, so the agreement between the identity and the simulated displacement rests on the feasible cells of the June 2026 runs, which are few at the eight-visit design and at 100 replicates there; and whether F1 recovers the slope at small $c_{bm,PB}$ on the sixteen-visit hybrid design is not yet known. Second, the η_t multipliers used to parameterize the expectancy modulation of PB are modeling assumptions, and a sensitivity analysis varying η across plausible values, which the protocol called for, was not run; no clinical conclusion should be drawn from the framework's outputs before it is. Third, the framework as specified treats the three components as monotone Gompertz trajectories; cyclical or biphasic TV patterns require either flexible TV specifications or explicit covariates beyond the Gompertz form, and the boundary between 'flexible-Gompertz' and 'covariate-augmented' specifications is not yet sharply drawn. Fourth, the participant-level component estimates the framework targets inherit the identifiability caveat that[56] and[51] raise for individual treatment effects more generally: within-participant component recovery rests on separating a true participant-by-component signal from participant-by-period and participant-by-phase variation, and this separation is credible only when the design supplies repeated phase contrasts and the cross-component covariance structure is not pathological. The framework should be read as estimating these quantities under those assumptions, not as dissolving the well-known difficulty of estimating individual effects from finite within-person data.

The companion pedagogical document at `docs/24-component-decomposition-pedagogy.md` develops the framework with worked examples and an extended intuition narrative for readers approaching the material for the first time. The companion methodological evaluation of

the Gompertz family at `analysis/report/07-gompertz-evaluation/` quantifies the sensitivity of the framework's inference to the specific parametric form chosen for the component trajectories.

A final point concerns the scope of the decomposition, where it is essential to distinguish the existence of the components from their identifiability. The claim that an observed longitudinal treatment response is the sum of a pharmacological, a belief-driven, and a natural-history component is a statement about the data-generating process of any repeated-measures treatment study, and in that sense the decomposition is not specific to N-of-1 trials. Identifiability, by contrast, is design-specific, and the distinction between population-level and participant-level recovery is decisive for the biomarker application. A standard parallel-group placebo-controlled trial supplies a between-arm placebo contrast that identifies the population-mean belief response but not a participant-level PB, and it contains no within-subject contrast in which belief varies independently of pharmacology; the participant-level biomarker-by-BR slope that the precision-medicine question targets is therefore not identified from such a design, no matter how large it is. The designs that do supply the required within-subject belief contrast are the blinded or open-hidden discontinuation and randomized-withdrawal designs[53,54], the crossover with on/off periods, and the hybrid N-of-1 design developed here; the pharmacometric disease-progression-plus-placebo models from which the framework descends[1–3] operate at the group level and recover population-mean components only. We therefore advocate the decomposition as a default lens for longitudinal designs that supply a within-subject belief-by-exposure contrast, with the aggregated N-of-1 case the most demanding instance because it must recover the components from a single participant's trajectory. We do not claim participant-level identifiability from parallel-group placebo control alone, and where only a between-arm placebo contrast is available the framework's claims are restricted to the population mean.

10 Conclusions

In conclusion, three points are to be emphasized. First, treatment response in aggregated N-of-1 clinical trials is the sum of three causally distinct components: pharmacological action (BR), placebo-belief response (PB), and time-variant natural history (TV). Each component has its own biological mechanism, its own identifiability conditions, and its own clinical interpretation. Whether conflating them biases a given analysis is, however, governed by the omitted-variable-bias identity of section 6.1, not by the raw magnitude of PB and TV: the one-component pre-post treatment effect is displaced by the PB and TV contributions, whereas the biomarker-treatment slope is displaced only by the biomarker's covariance with those components. Study 1 bears this out: varying PB and TV magnitude with a biomarker correlated with BR alone left the one-component estimate unbiased across the entire grid.

Second, the operative requirement is the design contrast, not the parametric analysis model. The trial design must include the drug-on/off and belief contrasts that identify the components, and the one-component analysis already exploits the drug-on/off contrast through its D_{bc} term. Adding parametric component structure on top is not free: under an uncontami-

nated biomarker the phase-augmented analysis reduced no bias and lost substantial power, so it should not be adopted as a default. Its value, and that of the full component-matched fit, is confined to the estimands where the extra structure does work, namely the attribution of response to pharmacology versus belief and the analysis of biomarkers that may themselves correlate with PB or TV.

Third, for predictive-biomarker validation the decomposition is needed precisely when the candidate biomarker may correlate with PB or TV. A biomarker that predicts the one-component total is a pharmacological predictor only when it is uncorrelated with the non-pharmacological components. Where that cannot be assumed, the one-component slope is displaced by $-c_{bm,PB}\,\sigma_{PB}\,(\bar{\eta}_{\text{on}} - \bar{\eta}_{\text{off}})$, a quantity the design fixes and the feasible cells of Studies 2 to 4 reproduce in sign and approximate size. Study 4 establishes that the displacement is removable: the F1 model returns the pharmacological slope across the feasible contamination range on a balanced-placebo design that varies drug exposure and belief separately. On the sixteen-visit hybrid design F1 is unbiased without contamination, and its behavior at the small contamination levels that design can accommodate awaits the corrected run. The practical lesson is therefore as much about design as about analysis, and the designs that supply the needed contrast require concealment that must be justified.

11 Conflict of interest

The authors declare no conflicts of interest.

12 Funding

This work received no specific funding.

13 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

14 Reproducibility

The simulations were run in a host R 4.4 session with the package loaded by `pkgload::load_all()`, as the driver headers state, not inside the project's Docker container; the manuscript is rendered through `tools/render.sh`, which stamps the rendering environment on each page.

The package set is captured in `renv.lock`. The analysis pipeline uses `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (`mclapply` over replicates), `data.table`, and `rmarkdown` with `bookdown` for the manuscript build.

Every driver seeds each replicate as `set.seed(study_seed + 1000L * cell_id + rep_idx)` with the study seed of Table 2: 642 for Study 1 (`run-study-a-prod.R`), 644 for Study 2 (`run-study-contam.R`), 815 for Study 3 (`run-study-b-recovery.R`), 916 for Study 4 (`run-study-b-balanced.R`) and 2061 for the corrected study (`run-study-2-feasible.R`). No driver calls `parallel::mc.set.seed`. All drivers are under `analysis/scripts/component-decomposition/` together with the feasibility script `05-feasibility-bounds.R`, the working ADEMP protocol and the deviations log. Per-cell checkpoints holding every replicate, the collated replicate files and the cell-level summaries are under `analysis/data/derived_data/component-decomposition/` in the directories of Table 2; the pilot alone is under `analysis/data/quick-sim/`. The stored `N` column of Studies 1 to 3 was recoded in place on 2026-08-08 (`migrate-n-to-total.R`) from the per-path count the drivers originally recorded to the total across paths, without a rerun; the draws are unchanged.

The June 2026 drivers of Studies 2 to 4 passed `makePositiveDefinite = TRUE` to `generateData()`, which at that time repaired a non-positive-definite covariance matrix silently. Section 7.1 identifies the cells this affected. No cell reported as a contamination result in this paper drew from a repaired matrix; the cells that did are reported only in Table 7. The corrected driver generates data under the package's current default, which refuses such a matrix, and never enables the repair. The manuscript source, the paper-specific bibliography (`references.bib`) and the CSL file are at `analysis/report/06-component-decomposition/`; the audit trail of corrections to earlier versions is in that directory's `README.md`.

Glossary

- **Biomarker-response correlation (c_{bm}).** The correlation between the baseline biomarker and the drug-response component at on-drug visits, 0.45 throughout; it decays off drug at the carryover half-life. The interaction is encoded through it (the correlation encoding), and the pharmacological slope is $-c_{bm}\,\sigma_{BR}/\sigma_{bm}$.
- **Biomarker-placebo correlation ($c_{bm,PB}$).** The correlation between the biomarker and the placebo-response component at every visit; the contamination parameter of Studies 2 to 4. Stored as `c_bm_pb`.
- **Feasibility bound.** The largest $c_{bm,PB}$ at which the joint covariance matrix is positive definite for a given c_{bm} and design, from the Schur complement on the biomarker row.
- **Repaired matrix.** A non-positive-definite covariance matrix replaced by a nearby positive definite one with `corpcor::make.positive.definite`; the June 2026 package did this without warning, the current package refuses instead.
- **Expectancy (η_t).** The participant's design-determined belief that treatment is active at visit t, the multiplier on the mean and standard deviation of the placebo-response component (Table 1). Stored as `De`.

- **Expectancy loading (w_{PB}).** The design's mean expectancy over on-drug visits minus its mean over off-drug visits, binary drug state.
- **One-component analysis.** `Sx ~ bm + t + Dbc + bm:Dbc` with random intercept and `corCAR1` residual correlation. Stored as `one_component` or `lumped`.
- **Phase-augmented analysis.** `Sx ~ bm + t + Dbc + phase + bm:Dbc + bm:Dbc:phase`, the identified part of the protocol's phase-augmented formula (section 5.2). Stored as `phase_augmented`.
- **F1.** `Sx ~ bm + t + Dbc + De + bm:Dbc + bm:De`, the series' recommended interaction analysis, called the belief-covariate decomposition in earlier versions. Stored as `decomposition` or `f1`.
- **Blinded-stratum analysis.** The one-component formula fitted to the visits with $\eta_t < 1$ alone. Stored as `decomposition_blinded`.
- **Eight-visit design.** Single-path open-label (five visits, on drug) then blinded discontinuation (three visits, on, off, off), 2.5-week spacing. Stored as `simple`.
- **Sixteen-visit design.** Two-path open-label (eight visits), blinded discontinuation (four visits, off drug), then a crossover of two two-visit periods, 1.25-week spacing. Stored as `full`.
- **Balanced-placebo design.** Three open-label on-drug visits, six covert on-drug visits ($\eta = 0$), six concealed-placebo visits ($\eta = 1$), 1.5-week spacing. Stored as `balanced` or `BAL`. Its covert visits are stored as `COV` and its concealed-placebo visits as `OLP`.
- **Hybrid design.** Open-label titration, blinded discontinuation, then a brief crossover; the eight-visit and sixteen-visit designs are single-path and two-path variants of it.
- $D_{bc,it}$**.** The exposure-decayed drug indicator of the analysis, built at the assumed half-life (1 week here). Stored as `Dbc`.
- **MCSE.** Monte Carlo standard error of a performance measure.

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